Gym Management Suite for Power World Gyms – Kiribathgoda

By

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2025/10/19

Declaration

I hereby certify that this project, titled **Gym Management Suite**, and all related artifacts are entirely my own work. It has not been submitted previously, nor is it being submitted concurrently, for any other degree or academic qualification at any institution.

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Abstract

Managing a fitness center involves a variety of daily operations that are essential for smooth workflow and high-quality member services. At Power World Gyms, Kiribathgoda, tasks such as attendance tracking, membership management, trainer scheduling, session coordination, and payment processing are currently performed manually. This traditional approach can be time-consuming, prone to errors, and limits the ability to provide personalized services, highlighting the need for a modernized management system.

The Gym Management Suite was developed to address these challenges by automating and streamlining the day-to-day operations of the gym. The system provides a unified platform for staff, trainers, and members, allowing accurate monitoring of attendance, efficient management of memberships, organized session scheduling, and timely tracking of payments. By centralizing these operations, the system ensures consistency, reduces administrative workload, and minimizes errors.

Beyond administrative efficiency, the system enhances the overall member experience. Trainers can manage their schedules effectively, monitor member progress, and provide personalized workout guidance, while members benefit from streamlined registration, clear session information, and timely updates about their memberships and workouts. This creates a more organized, responsive, and engaging environment for all stakeholders.

The system also supports the gym management team in making informed decisions by providing clear insights into operational trends, member engagement, and financial tracking. By modernizing previous manual processes, the system allows staff to focus on improving service quality and offering better fitness experience rather than being burdened by routine administrative tasks.

This report presents a comprehensive analysis of the proposed system, covering its objectives, design, and functionality, as well as the practical benefits it offers. The Gym Management System demonstrates how automation and structured workflows can transform daily operations, improve efficiency, and contribute to enhanced member satisfaction at Power World Gyms.

Chapter 1 – Current System Analysis

Introduction

This chapter provides an overview of the gym's current operations, highlighting the processes, challenges, and inefficiencies faced in daily management. It also outlines the goals, potential improvements, and the scope of the proposed system, setting the foundation for understanding how operations can be optimized and modernized.

Business Overview

Power World Gym in Kiribathgoda, located at 311 Kandy Road, Gala Junction, is a branch of Sri Lanka's largest fitness chain. Founded in 1994, Power World pioneered the concept of making fitness affordable and accessible to the country's middle-income population. The Kiribathgoda facility is part of a 24-branch network and embodies this mission by providing comprehensive, functional workout experience. It serves a diverse client base, from beginners to seasoned athletes, offering the essential equipment and environment needed to achieve personal health goals in a straightforward, value-focused setting.

The gym operates with extensive hours, from 5:00 AM to 10:00 PM, Monday through Saturday. Inside the facility members find a wide selection of modern cardio and strength-training machines, alongside a variety of free weights. Trainers are present to offer assistance and guidance when requested, aligning with a model that supports self-directed exercise. Services such as personalized training plans and nutritional advice are available to help members on their fitness journey, making it a well-rounded facility for the local community.

Figure 1: Exterior view of the facility

Figure 3: Interior view of the facility – first floor

Figure 2: Interior view of the facility – second floor

Figure 4: QR Code notice pasted at the front door

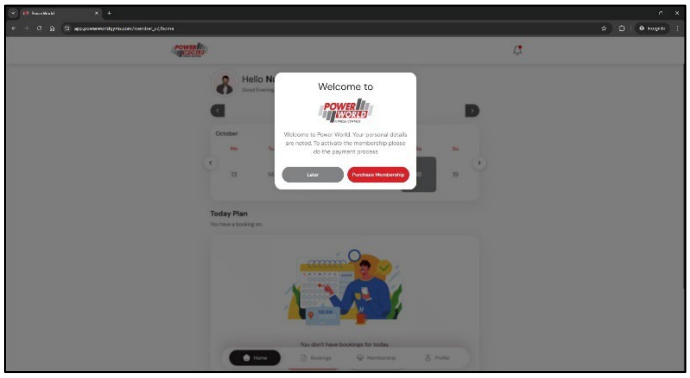


Figure 5: Current system handling only payments

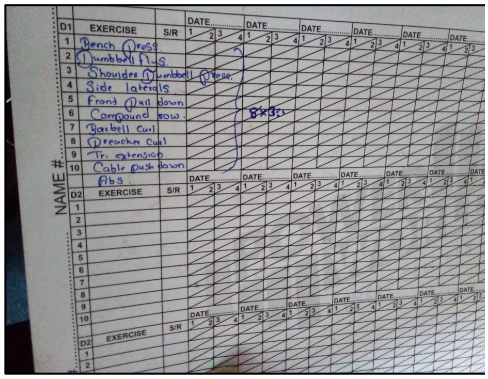


Figure 6: Workout schedule card given

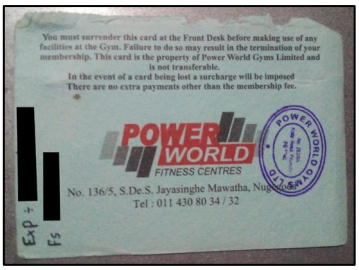


Figure 7: Membership card

Business Process

Currently, the core operational processes at the Gym are managed through manual and semi-digital methods. The staff handles everything from member registration and attendance logging to payment processing and workout schedule coordination. This traditional approach creates bottlenecks and limits the gym's ability to offer personalized, efficient service. The main processes that will be focused on are:

1. Member Registration and Onboarding
2. Subscription and Payment Management
3. Gym Access and Attendance Tracking
4. Workout Planning and Progress Tracking
5. Instructor Scheduling and Member Communication

A brief description of each of these processes is explained below.

1. Member Registration and Onboarding

New members must register on-site by filling out paper forms. Staff manually issue membership cards and record personal details. This process is slow, does not facilitate immediate digital profile creation, and can lead to data entry errors or lost forms.

2. Subscription and Payment Management

While a QR code system exists, it only redirects to a basic website for subscription payments. It does not integrate with other gym systems for automated access control or personalized service activation. Cash payments are also accepted, requiring manual settlement.

3. Gym Access and Attendance Tracking

Members sign a physical attendance book upon entry and exit. This method is prone to errors, fraudulent sign-ins, and provides no real-time data on gym occupancy. Records are unavailable outside operating hours and are difficult to analyze for trends.

4. Workout Planning and Progress Tracking

Workout routines are often generic (non-personalized routines) and not tailored to individual member goals. Instructors provide schedules manually upon request, but there is no centralized

system to track a member's adherence, progress, or update their plan based on performance. Manual entry of vitals like BMI is inconsistent.

5. **Instructor Scheduling and Member Communication**

Instructors operate with significant independence, leading to inconsistencies in availability and support. New members often feel unsupported as there is no formal system for assigning a trainer or facilitating communication between members and staff.

Problems and Weaknesses

1. **Time-Consuming Operations:** Manual registration, attendance-taking, and payment processing consume valuable staff time that could be dedicated to member service.

2. **High Risk of Human Error:** Handwritten records for attendance and member details are prone to inaccuracies and can be easily lost or damaged.

3. **Ineffective Member Tracking:** The inability to consistently track member attendance, workout completion, and vital statistics (BMI, weight) prevents personalized service and progress monitoring.

4. **Poor Communication and Support:** The lack of a unified communication platform leads to delays in addressing member inquiries and inconsistent trainer support, diminishing the member experience.

5. **Limited and Siloed Data:** The existing QR payment system operates in isolation. The gym lacks a central database, making it impossible to generate insightful reports on membership trends, financial performance, or member engagement.

6. **Lack of Personalization:** The absence of AI-driven features and digital profile management means members receive generic workout plans and no adaptive guidance, meal suggestions, or automated reminders.

Aims and Objectives

Aim

This project aims to transform Power World Gym into a modern, efficient, and member-centric facility by implementing a comprehensive digital Gym Management Suite. This system will automate daily operations, enhance communication, and provide data-driven insights for strategic growth.

Objectives

1. **Automate Member Lifecycle Management:** The system will digitize the entire member journey, from online registration and profile creation to automated subscription management and digital check-ins. This will reduce administrative workload, minimize errors, and provide seamless onboarding experience.

2. **Enhance Member Experience and Engagement:** Through a dedicated mobile app, members will receive personalized workout plans, have access to an AI assistant for queries and meal planning, and be able to track their progress seamlessly. This creates a supportive and tailored fitness journey, increasing member retention.

3. **Streamline Instructor and Staff Workflows:** Instructors will have a mobile app to manage their schedules, view assigned members' progress and communicate directly. The staff web portal will provide tools for routine building, shift coordination, and a comprehensive overview of all gym operations, ensuring accountability and efficiency.

4. **Implement Centralized Data and Reporting:** All member, financial, and operational data will be consolidated into a single system. This will enable the generation of automatic reports on revenue, attendance, and member progress. AI-driven analytics will provide forecasts for membership growth and identify operational risks.

5. **Improve Financial Transparency and Control:** The system will centralize all payment processing, providing clear tracking of subscriptions, instructor payouts, and overall revenue. This ensures accurate financial records and simplifies reconciliation.

6. **Ensure System Scalability and Future-Proofing:** Built with a modular architecture, the system will be able to grow with the gym. New features can be added without disrupting core operations, supporting the long-term expansion of the business.

Scope of the Project

The aims of this project are to design and develop a web-based system for efficiently managing the gym's daily operations and member activities. The system focuses on automating key management tasks and improving user experience. In the future, dedicated mobile applications will be introduced to enhance accessibility for both members and staff.

Main features of such a system can be listed as below:

Registration and Profile Management: For members, instructors, and staff. Members can register online, manage their profiles, and track goals. Instructors and staff have role-based access to manage schedules and member information.

Subscription and Payment Gateway: A secure, integrated system for managing one-time and recurring payments, directly linked to gym access and service activation.

Access Control and Attendance Tracking: Digital check-in/check-out using QR codes scanned via the member app, providing real-time attendance logs and access control.

Workout and Service Management: Tools for creating, assigning, and modifying personalized workout routines. A scheduling system for booking sessions with instructors.

AI-Powered Insights and Support: An AI chatbot for member inquiries, adaptive workout and meal suggestions, and predictive analytics for business intelligence on the staff portal.

Comprehensive Reporting: Generation of detailed reports on membership, attendance, financial performance, and member progress.

User Stories

The following user stories describe the specific goals, needs, and motivations of the four primary user groups: Member, Trainer, Manager, and Admin. These narratives translate high-level system capabilities into tangible user benefits, forming a critical foundation for defining detailed functional requirements and ensuring the final solution delivers meaningful value to all stakeholders.

Member User Stories

Id	**User Story**
US 01	As a member, I want to scan a QR code to get into the gym, so that I don't have to manually check in with a person or a book.
US 02	As a member, I want to track my progress over time by seeing my workout history, so I can stay motivated.
US 03	As a member, I want to make an appointment with a trainer through an app, so that it's fast and I can see their real-time availability.
US 04	As a member, I want to buy a subscription online, so that I can start using the gym immediately.
US 05	As a member, I want to mark my workout as done, so that I can keep a record of my completed sessions.
US 06	As a member, I want to receive a notification if my subscription is about to expire, so that I can renew it and avoid any interruption.
US 07	As a member, I want to view insights about my own attendance and workout frequency, so I can see my own consistency.
US 08	As a member, I want to modify my workout schedule in the app if I can't make it, so that I can manage my time effectively.

Trainer User Stories

Id	User Story
US 09	As a trainer, I want to see which of my clients have been consistently recording their workouts, so I know who is following the plan.
US 10	As a trainer, I want to get a notification when a member makes an appointment with me, so I am prepared for their session.
US 11	As a trainer, I want to modify workout schedules for my clients, so their plans stay challenging and effective.
US 12	As a trainer, I want to set my own available hours in the system, so that members can only book sessions when I am working.
US 13	As a trainer, I want to access detailed reports on a client's past sessions and progress, so I can provide better guidance.
US 14	As a trainer, I want to view insights on my clients' overall performance, so I can easily identify who needs more attention.
US 15	As a trainer, I want to see a member's completed workouts before they come in, so I can track their effort between our sessions.

Manager User Stories

Id	User Story
US 16	As a manager, I want to view insights on membership growth trends so that I can plan future business strategies.
US 17	As a manager, I want to access detailed reports on revenue and subscription types to understand what is driving our profits.
US 18	As a manager, I want to see reports on gym usage patterns to make decisions about operational hours and staff allocation.
US 19	As a manager, I want to identify members with declining attendance, so I can understand potential risks to member retention.
US 20	As a manager, I want to see which trainers have the most appointments booked, so I can recognize high-performing staff.

US 21	As a manager, I want a simple dashboard that shows me today's revenue, total active members, and gym occupancy at a glance, so I don't have to dig through spreadsheets.
US 22	As a manager, I want to forecast future membership numbers based on current data, so I can prepare for growth.

Admin User Stories

Id	User Story
US 23	As an admin, I want to create and configure new subscription plans, so that we can offer different membership tiers.
US 24	As an admin, I want there to be a 'Forgot Password' feature for members, so that I am not involved in simple account recovery.
US 25	As an admin, I want to monitor the status of the QR code scanners and door locks, so that I can ensure the entrance system is always operational.
US 26	As an admin, I want to receive a system alert if the payment gateway is down, so that I can resolve the issue and prevent failed transactions.
US 27	As an admin, I want to define user roles and permissions for staff and trainers, so that data security and access are properly controlled.
US 28	As an admin, I want to be able to run system backups, so that all member and business data is secure.
US 29	As an admin, I want to manage the entire system by configuring system settings and operational limits, so that the gym's business rules are enforced automatically.

Diagrams

Use-Case Diagram

This section presents an overall use case diagram illustrating the main interactions between the key actors, including members, trainers and administrators, along with the processes of the existing manual system. The diagram highlights the major activities performed by each actor and provides a visual summary of the system's current functionality and its limitations.

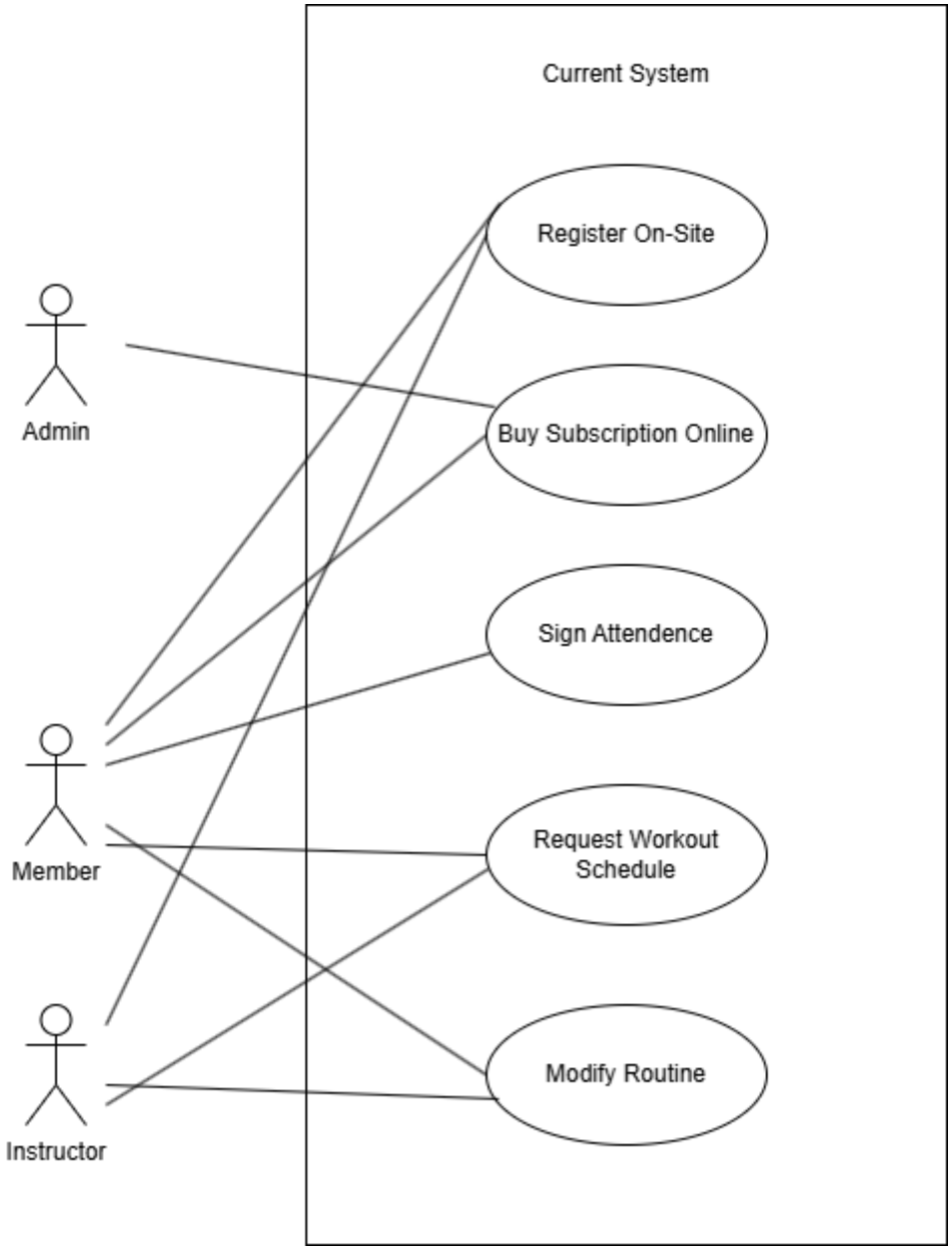


Figure 1-1: Use Case Diagram for Existing System

Use-Case Descriptors

Use Case: Register On-Site

Id	CUC-01
Name	Register On-Site
Actors	Member (Primary), Instructor (Secondary)
Description	This use case describes the process where a new member registers for the gym in person by providing their details, signing a form, and making a cash payment to an instructor, who then issues a physical membership card.
Pre-Conditions	• The prospective member is physically present at the gym. • The instructor is available at the gym.
Post-Conditions	• A new member record is created. • A physical membership card is issued to the new member. • The member is officially registered and can use gym facilities.
Main Flow	1. The Member enters the gym. 2. The Instructor provides a paper registration form. 3. The Member fills it in. 4. The Member signs and pays. 5. The Instructor issues the card.
Alternative Flows	A1: Invalid Form Details – Instructor requests corrections before payment.

Use Case: Buy Subscription Online

Id	CUC-02
Name	Buy Subscription Online
Actors	Member (Primary), System (Secondary)
Description	This use case describes the process where an existing member purchases or renews a subscription plan through the gym's website.
Pre-Conditions	• Member has internet access. • Member has a valid online payment method.
Post-Conditions	• Member's subscription is activated or renewed. • Confirmation message is recorded and sent.
Main Flow	1. Member scans QR code. 2. System displays plans. 3. Member selects plan and enters details. 4. System processes payment. 5. System activates subscription. 6. Confirmation sent to Member.
Alternative Flows	A1: Payment Failure – System prompts to retry payment.

Use Case: Sign Attendance

Id	CUC-03
Name	Sign Attendance
Actors	Member (Primary), Instructor (Secondary)
Description	This use case describes the manual process of a member signing a physical attendance book upon entering the gym, with the instructor verifying the entry.
Pre-Conditions	• Member is registered. • Member is physically present. • Attendance book available.
Post-Conditions	• Attendance recorded and verified by Instructor.
Main Flow	1. Member enters gym. 2. Member presents attendance book. 3. Instructor marks attendance. 4. Member signs book. 5. Instructor verifies record.
Alternative Flows	None specified.

Use Case: Request Workout Schedule

Id	CUC-04
Name	Request Workout Schedule
Actors	Member (Primary), Instructor (Secondary)
Description	This use case describes the process where a member, after discussing their workout history and goals with an instructor, receives a new, personalized workout schedule.

Pre-Conditions	• Member is registered and active. • Both are present for a meeting.
Post-Conditions	• Member's needs documented. • New schedule provided.
Main Flow	1. Member meets Instructor. 2. Member informs completion or goal change. 3. Member requests new schedule. 4. Member provides requirements. 5. Instructor gives detailed schedule.
Alternative Flows	A1: Member continues own routine – Use case ends.

Use Case: Modify Routine

Id	CUC-05
Name	Modify Routine
Actors	Member (Primary), Instructor (Secondary)
Description	This use case describes the process where a member meets an instructor to review and update their existing workout routine based on progress or changing needs.
Pre-Conditions	• Member has an existing routine. • Both present for a meeting.
Post-Conditions	• Routine reviewed. • Routine updated.
Main Flow	1. Member meets Instructor. 2. Instructor reviews routine.

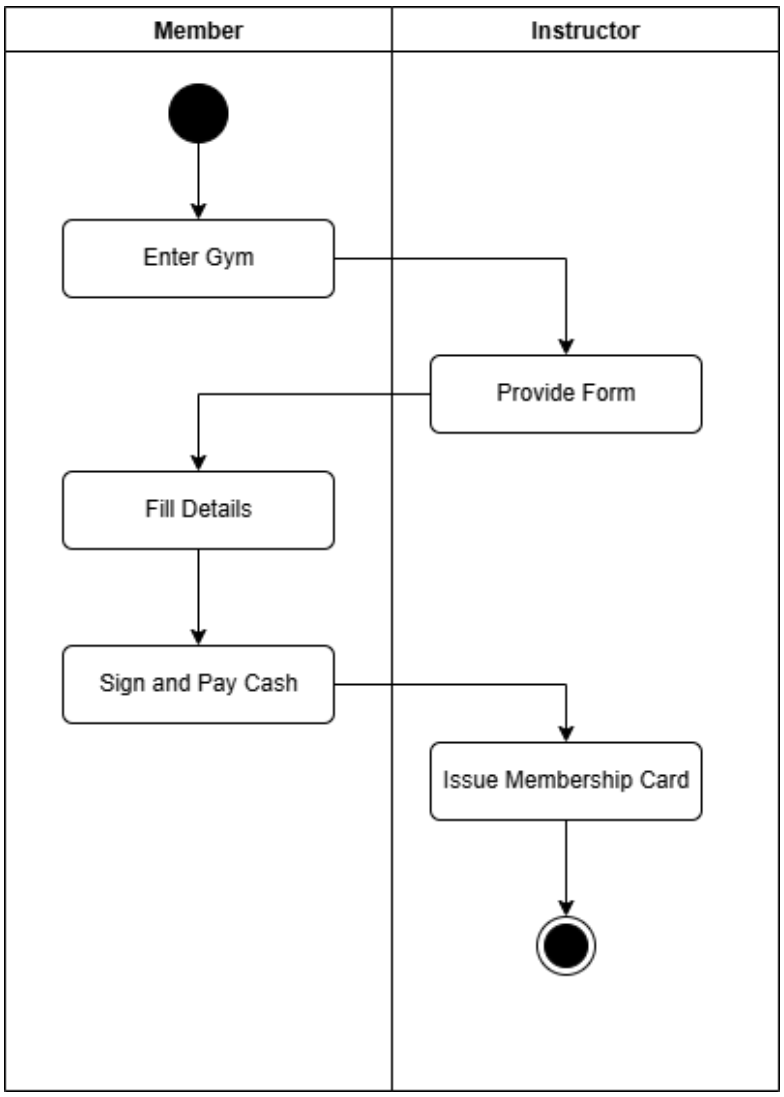
	3. Instructor suggests changes. 4. Member accepts changes. 5. Instructor updates routine.
Alternative Flows	A1: Member rejects changes – Instructor may propose alternatives or end use case.

Activity Diagrams

Activity diagrams are used here to visually represent the workflows of the existing manual processes. Each diagram outlines the sequence of actions and decisions carried out by different actors, showing how tasks progress from one stage to another. By mapping these steps, the diagrams provide a clear understanding of how activities are currently performed, the interactions between participants, and the decision points that guide the overall flow of operations.

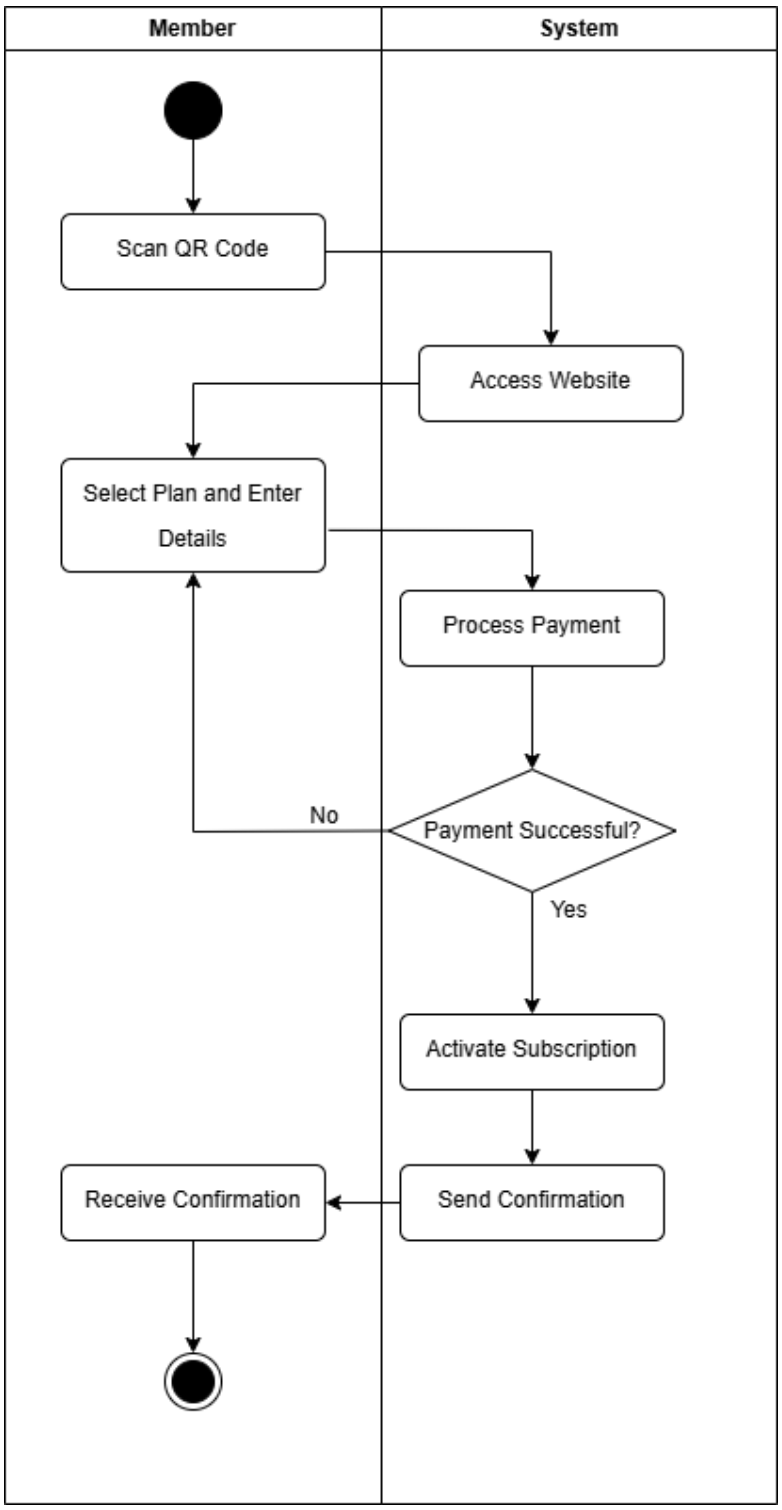
Activity: Register On-Site

A member enters the gym, receives a form from an instructor, fills in their details, and pays in cash. The instructor then issues a physical membership card to complete the enrollment process.



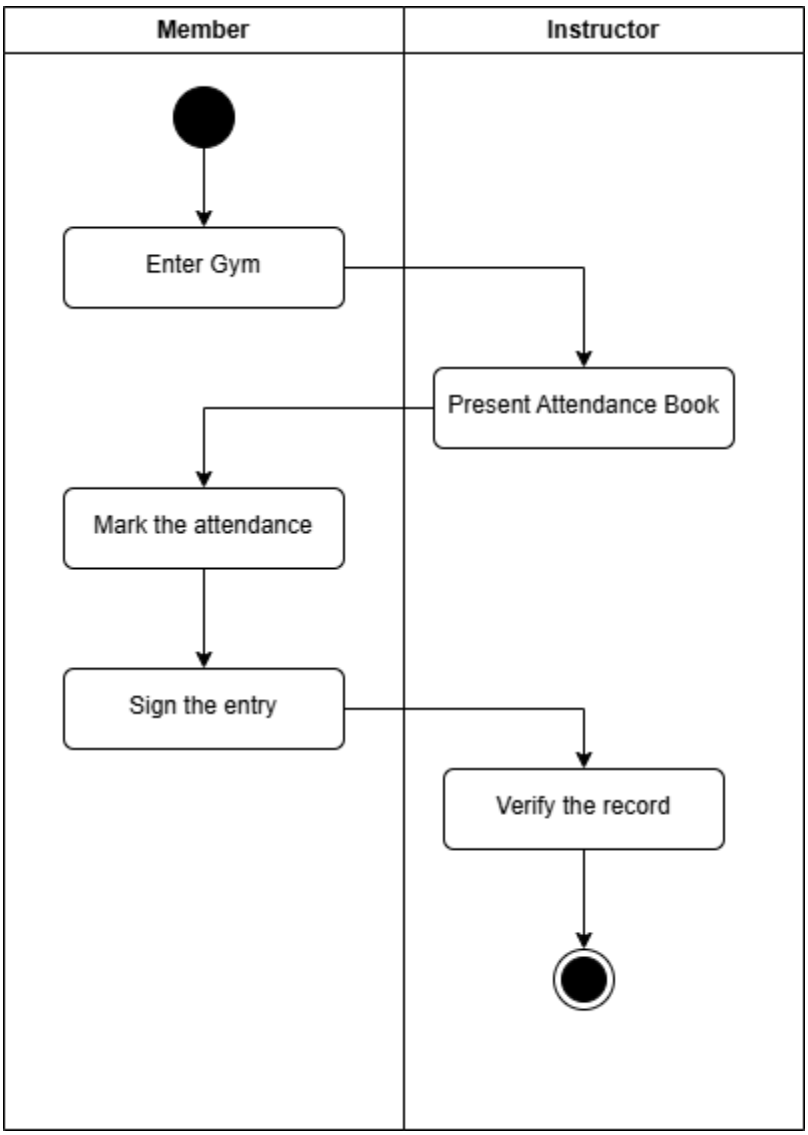
Activity: Buy Subscription

A member scans a QR code to access the gym's website, selects a plan, and enters their details. The system processes the payment and, upon success, activates the subscription and sends a confirmation to the member.



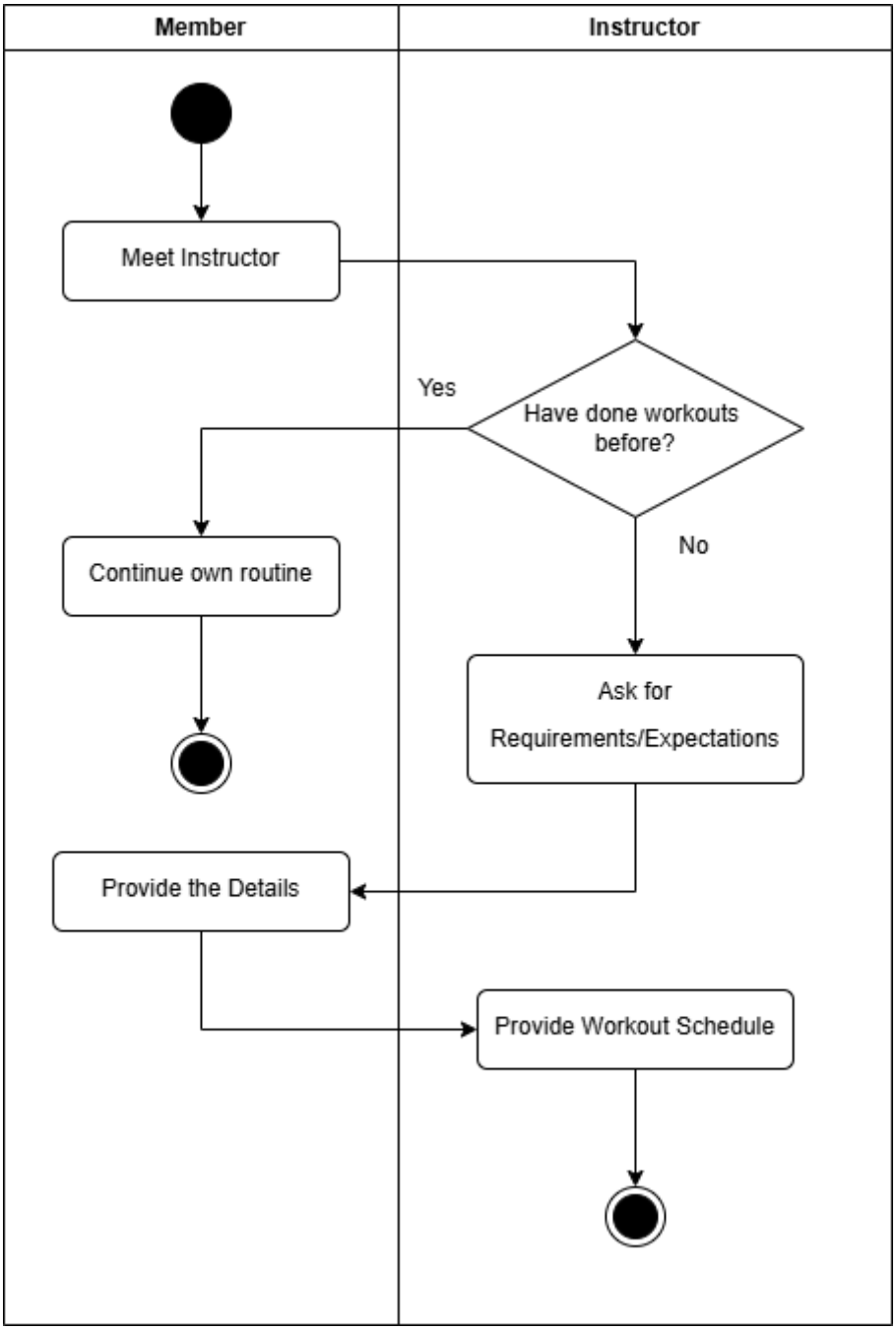
Activity: Sign Attendance

A member presents a physical attendance book to an instructor at the gym. The instructor marks the attendance, the member signs the entry, and the instructor verifies the record.



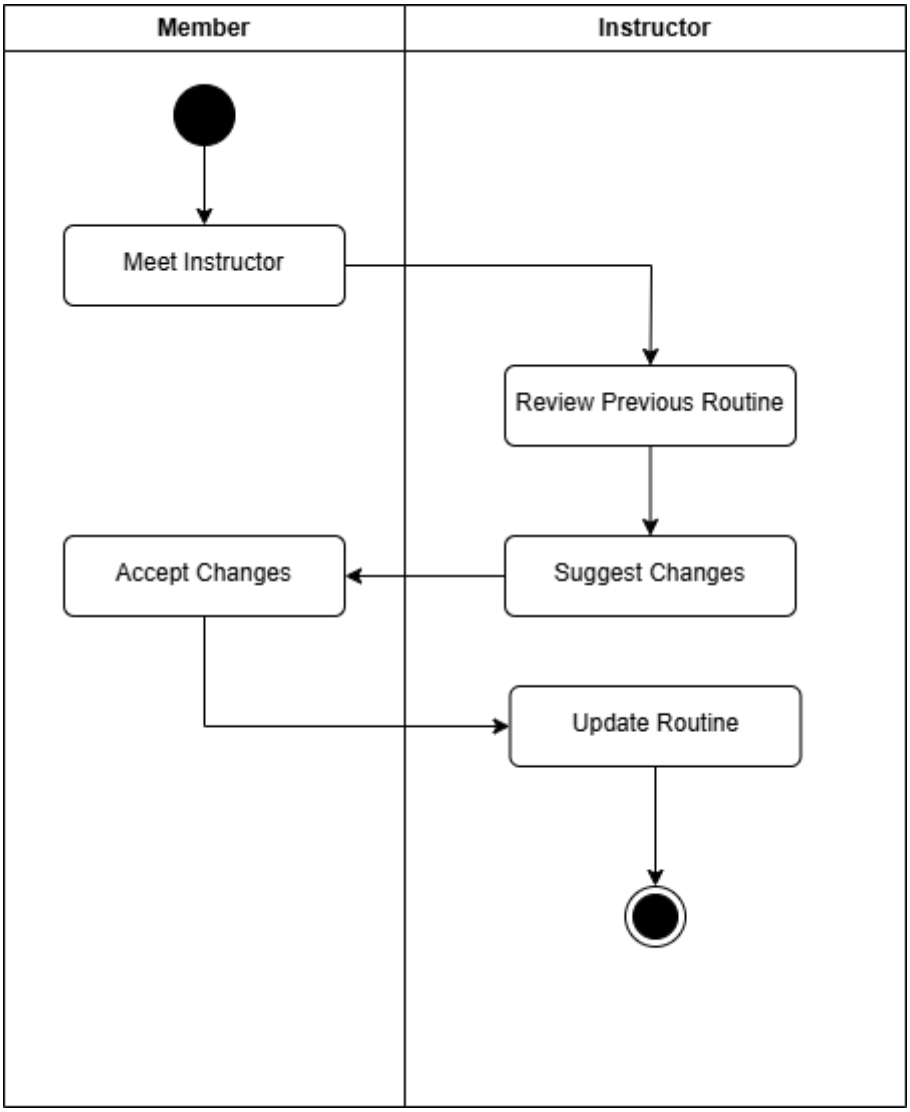
Activity: Request Workout Schedule

A member meets with an instructor, discusses their workout routine, and shares their requirements and expectations. The instructor then uses this information to provide a personalized workout schedule.



Activity: Modify Routine

A member and instructor meet to review the member's previous workout routine. The instructor suggests changes, and after the member accepts them, the routine is officially updated.



Chapter 2 – Proposed System Analysis

Introduction

The proposed Gym Management System is a comprehensive digital solution designed to replace the existing manual and semi-digital processes at Power World. The current system reliant on physical cards, paper-based logs, and static schedules is inefficient, prone to error, and limits member engagement and administrative oversight, as discussed in the previous chapter.

This new integrated system aims to streamline all core gym operations into a single, secure, and user-friendly platform. Its primary objective is to automate key processes such as member access, attendance tracking, subscription management, and appointment scheduling. By digitizing these workflows, the system enhances operational accuracy, improves security, enriches the member experience through self-service features and progress tracking, and provides management with valuable, data-driven insights for strategic decision-making. The system is designed to be scalable, reliable, and accessible via various endpoints like mobile devices, and web portals.

Software Requirements Specification

This segment presents the detailed Requirements Specification for the new Gym Management System. It consolidates all necessary criteria derived from user stories and documented use cases, forming a complete blueprint for development.

This compiled list acts as a definitive checklist, outlining the essential functions the system must perform. It also defines the critical performance benchmarks and usability standards it must achieve to be effective and user-friendly for all stakeholders.

These requirements are primarily divided into,

1. **functional requirements** and
2. **non-functional requirements**

Functional Requirements

Functional requirements specify the core operations and feature a system must deliver to meet its intended purpose. They define what the system is expected to do and outline the services it should provide to users.

These requirements typically address:

1. System Capabilities – The essential functions and tools the system must support.
2. User Interactions – The ways in which users engage with the system, including inputs, outputs, and interface behavior.
3. Data Management – How the system processes, stores, retrieves, and secures information.

ID	Requirement Description	Priority	Type
FR-01	The system shall allow a member to scan a QR code using a mobile device for identification.	High	Mandatory
FR-02	The system shall validate the member's subscription status in real-time upon a scan attempt.	High	Mandatory
FR-03	The system shall authenticate the user's identity based on the scanned QR code data.	High	Mandatory
FR-04	Upon successful authentication and validation, the system shall send a signal to unlock the entrance door.	High	Mandatory
FR-05	The system shall record a timestamp with member ID, event type (IN/OUT), and location for every entry and exit.	High	Mandatory
FR-06	The system shall prevent duplicate "IN" events for a member who is already logged inside.	Medium	Mandatory
FR-07	The system shall provide instructive error messages for failed scans, invalid subscriptions, or authentication failures.	Medium	Mandatory
FR-08	The system shall allow staff members to enter/leave the facility, logging their access.	Medium	Mandatory
FR-09	The system shall allow a member to purchase a new subscription plan via the web or mobile app.	High	Mandatory

FR-10	The system shall integrate with a payment gateway to process subscription payments.	High	Mandatory
FR-11	Upon successful payment, the system shall automatically activate the member's subscription and update their record.	High	Mandatory
FR-12	The system shall send a confirmation notification (e.g., email) to the member after a successful purchase.	Medium	Mandatory
FR-13	The system shall allow members to view available time slots and instructors for appointments.	Medium	Mandatory
FR-14	The system shall allow a member to book an appointment with a selected instructor for an available time slot.	Medium	Mandatory
FR-15	The system shall verify a member's subscription entitlements before allowing an appointment to be booked.	Medium	Mandatory
FR-16	The system shall send notifications to both the member and the instructor upon appointment creation.	Medium	Mandatory
FR-17	The system shall allow members and instructors to modify or reschedule existing workout appointments.	Medium	Mandatory
FR-18	The system shall check for scheduling conflicts (e.g., trainer availability) before confirming any modification.	Medium	Mandatory
FR-19	The system shall allow a member to mark a scheduled workout or routine as completed.	Medium	Mandatory
FR-20	The system shall record a completion timestamp when a workout is marked as done.	Medium	Mandatory
FR-21	The system shall recalculate and update the member's progress statistics upon workout completion.	Medium	Mandatory
FR-22	The system shall display a dashboard to the member with historical workout data, trends, and visualizations.	Medium	Mandatory

FR-23	The system shall provide an admin console for managing facilities, user roles, pricing plans, and system settings.	Low	Mandatory
FR-24	The system shall allow managers to generate and export detailed reports (e.g., attendance, revenue) in PDF/CSV formats.	Low	Mandatory
FR-25	The system shall validate user roles and permissions before allowing access to any functionality.	High	Mandatory
FR-26	The system should provide an insights dashboard for managers showing trends, occupancy patterns, and recommended actions.	Low	Optional
FR-27	The system should store data locally and sync later if a network connection is unavailable (offline capability).	Medium	Optional
FR-28	The system shall log all user and administrative actions for audit purposes.	Low	Mandatory

Non-Functional Requirements

Non-functional requirements define the standards and attributes that shape how a system performs, rather than the specific tasks it carries out. These requirements play a critical role in ensuring the system operates smoothly, efficiently, and remains dependable over time.

They typically emphasize:

1. Performance – How fast, responsive, and efficient the system is under various conditions.

2. Reliability & Availability – The system's ability to function consistently and remain accessible when needed.

3. Scalability & Maintainability – How well the system adapts to growth and how easily it can be updated or improved.

ID	Requirement Description	Priority	Type
NFR-01	The system shall use secure, signed, or one-time-use QR codes to prevent replay attacks.	High	Mandatory
NFR-02	The system shall implement rate-limiting on authentication attempts to prevent brute-force attacks.	High	Mandatory
NFR-03	All sensitive data (passwords, payment info) shall be encrypted in transit (TLS 1.2+) and at rest (AES-256).	High	Mandatory
NFR-04	The system shall respond to a QR scan and unlock door signal in under 2 seconds to prevent entrance delays.	High	Mandatory
NFR-05	The reporting module should generate standard reports for a 30-day period in under 10 seconds.	Medium	Optional
NFR-06	The core access control system shall have 99.9% uptime during gym operating hours.	High	Mandatory
NFR-07	The system should gracefully handle failure of a secondary service (e.g., if email service is down, queue notifications).	Medium	Optional
NFR-08	The system shall provide clear user interface messages and instructions.	Medium	Mandatory
NFR-09	The member mobile app should be learnable for a new user to perform basic tasks (check-in, view schedule) within 5 minutes.	Medium	Optional
NFR-10	The member mobile app shall be compatible with iOS (version XX and above) and Android (version YY and above).	High	Mandatory
NFR-11	The system shall integrate with standard payment gateways (e.g., Stripe, PayPal) using common APIs.	High	Mandatory

NFR-12	The system architecture should support a 20% increase in member count without a loss in performance.	Low	Optional
NFR-13	The system should be modular, allowing individual components (e.g., billing, auth) to be updated independently.	Low	Optional
NFR-14	The system shall validate event ordering to prevent illogical records (e.g., two consecutive "IN" events).	Medium	Mandatory

Business System Options (BSOs)

The following options specify alternative solutions to develop the system, with different functional scopes. Each option is evaluated based on its ability to meet user needs, fulfill business objectives, and enhance operational efficiency.

BSO 1: Essential Digitalization & Access Control

Description:

This option addresses the most critical flaw - the disconnect between online payments and gym access. It implements a mandatory QR-Based Access Control system. This forces digital attendance logging and, crucially, validates the online subscription at the door before granting entry. It replaces the physical sign-in book and ensures only paying members enter.

Solves Current Pain Points:

1. Eliminates the risk of members entering with expired/without subscriptions.
2. Automates and mandates attendance tracking, replacing the optional paper log.
3. Provides a definitive record of member visits.

Pros:

1. Highest Impact for Cost: Directly stops revenue leakage.
2. Faster Member Entry: Scans are quicker than manual sign-in.
3. Foundation for Future Systems: Establishes the core digital member database.

Cons:

1. Requires investment in hardware (Displays/tablets for entrance).
2. Members must adapt to using phones/QR codes instead of just walking in.

BSO 2: Operational Efficiency & Member Portal

Description:

This option builds on BSO 1 by digitizing the other major paper-based processes. It adds the Member Self-Service Portal featuring Trainer Appointments and Digital Workout Scheduling. This eliminates the outdated 'generic workout card' and allows members to book and manage their routines via an app, reducing constant instructor admin tasks.

Solves Current Pain Points:

1. Replaces physical workout cards with dynamic, updatable digital plans.
2. Streamlines the ad-hoc booking process, reducing scheduling conflicts and instructor workload.
3. Provides a clear value proposition to members (convenience), improving retention.

Pros:

1. Significantly reduces administrative burden on instructors.
2. Improves member satisfaction and engagement through modern self-service tools.
3. Creates a new structured data source (appointment history).

Cons:

1. Higher development cost than BSO 1.

BSO 3: Integrated Growth & Insights Platform (Recommended)

Description:

This is the complete system as proposed. It includes BSO 1 and 2 and adds the data analytics layer. This introduces Progress Tracking for members and Insights & Reporting for management. This turns operational data into strategic assets.

Solves Current Pain Points:

1. Replaces the non-existent manual metrics input (Such as BMI, weights) with a seamless digital progress journey, a powerful member retention tool.

2. Empowers 'Top Management' and 'Admin' with real-time insights into attendance trends, revenue, and facility usage, moving them from passive overseers to active managers.
3. Provides a complete 360° view of both business health and member engagement.

Pros:

1. Maximizes ROI: Uses data to drive membership retention, personalized marketing, and operational decisions (e.g., staff scheduling during peak hours).
2. Creates a Competitive Advantage: Offers features (progress analytics, insights) that differentiate the gym from competitors.
3. Future-Proofs the Business: The integrated platform can easily scale to add new services.

Cons:

1. Highest initial investment and complexity.
2. Requires the most cultural shift in how both staff and members interact with the gym.

BSO vs Functional Requirements

Id	Requirement	BSO 1	BSO 2	BSO 3
FR-01	The system shall allow a member to scan a QR code for identification.	X	X	X
FR-02	The system shall validate the member's subscription status in real-time.	X	X	X
FR-03	The system shall authenticate the user's identity based on the scanned QR code.	X	X	X
FR-04	The system shall unlock the entrance door upon successful authentication.	X	X	X
FR-05	The system shall record a timestamp for every member entry and exit.	X	X	X
FR-06	The system shall prevent duplicate entry events.	X	X	X
FR-07	The system shall provide instructive error messages for failed access attempts.	X	X	X
FR-08	The system shall allow and log staff member access to the facility.	X	X	X
FR-09	The system shall allow a member to purchase a subscription online.	X	X	X
FR-10	The system shall integrate with a payment gateway.	X	X	X
FR-11	The system shall automatically activate a subscription upon payment.	X	X	X
FR-12	The system shall send a confirmation upon purchase.	X	X	X
FR-13	The system shall allow members to view appointment availability.		X	X

FR-14	The system shall allow a member to book an appointment with a trainer.		X	X
FR-15	The system shall verify subscription entitlements before booking.		X	X
FR-16	The system shall send notifications upon appointment creation.		X	X
FR-17	The system shall allow modification of workout appointments.		X	X
FR-18	The system shall check for scheduling conflicts before modifications.		X	X
FR-19	The system shall allow a member to mark a workout as completed.		X	X
FR-20	The system shall record a timestamp for completed workouts.		X	X
FR-21	The system shall update member progress statistics.		X	X
FR-22	The system shall display a member progress dashboard.		X	X
FR-23	The system shall provide an admin console for system management.			X
FR-24	The system shall allow managers to generate detailed reports.			X
FR-25	The system shall validate user roles and permissions.	X	X	X
FR-26	The system should provide a manager insights dashboard.			X
FR-27	The system should operate with offline capability.			X
FR-28	The system shall log all user and administrative actions.	X	X	X

BSO vs Non-Functional Requirements

Id	Requirement	BSO 1	BSO 2	BSO 3
NFR-01	The system shall use secure QR codes to prevent replay attacks.	X	X	X
NFR-02	The system shall implement rate-limiting on authentication attempts.	X	X	X
NFR-03	The system shall encrypt all sensitive data in transit and at rest.	X	X	X
NFR-04	The system shall respond to a QR scan in under 2 seconds.	X	X	X
NFR-05	The system should generate standard reports in under 10 seconds.			X
NFR-06	The core system shall have high uptime during operating hours.	X	X	X
NFR-07	The system should gracefully handle secondary service failures.			X
NFR-08	The system shall provide a clear user interface.	X	X	X
NFR-09	The mobile app should be learnable for basic tasks within 5 minutes.		X	X
NFR-10	The mobile app shall be compatible with iOS and Android.	X	X	X
NFR-11	The system shall integrate with standard payment gateways.	X	X	X
NFR-12	The system should support a significant increase in members.			X
NFR-13	The system should be modular for independent component updates.			X
NFR-14	The system shall validate event ordering to ensure data integrity.	X	X	X

Selected BSO with Justification

Following a review of the options, the client has suggested that **BSO 3: Integrated Growth & Insights Platform** is the most suitable for the Power World Gym Management System.

This recommendation is based on its capacity to deliver a complete, strategic solution that addresses all identified limitations of the current manual processes. While BSO 1 provides a solid foundation for securing revenue and BSO 2 significantly enhances operational efficiency, BSO 3 is the only option that incorporates the tools necessary for long-term growth and market leadership. It not only automates core processes like access control and scheduling but also leverages the resulting data to enhance member retention through progress tracking and empower management with actionable business intelligence.

This option aligns with the client's comprehensive vision by providing a unified platform for all users. It transforms the gym from a traditional facility into a data-informed, modern fitness service. The platform's scalable design ensures it can grow alongside the business, and its integrated nature is expected to justify the initial investment through maximized operational efficiency, improved decision-making, and a superior member experience that fosters loyalty.

Feasibility Study

Technical Feasibility

- Assessment: The proposed system is Technically Feasible.
- Technology Availability: All core technologies required are mature, readily available, and well-understood.

- Front-end: Mobile apps (iOS/Android) can be developed with frameworks like React Native or Flutter. Web interfaces for admin and members can use standard JavaScript frameworks (React, Angular).
- Back-end: Standard cloud providers (AWS, Azure, Google Cloud) offer all necessary services (databases, serverless functions, authentication services).
- Hardware: Tablets or displays, and electronic door locks are commercial off-the-shelf (COTS) products with standard APIs for integration.

- Integration Complexity: The system requires integration between several modules (auth, billing, scheduling) and third-party services (payment gateways like Stripe). This presents a medium level of complexity but is a common practice with well-documented APIs and can be managed by a competent development team.

- Technical Skills Required: Development requires standard skills in full-stack web and mobile development, API design, and database management. These skills are widely available in the market. Maintenance post-launch requires a simpler skill set, manageable by a system administrator or a managed service provider.

- Scalability: The proposed cloud-based architecture is inherently scalable. Database and server resources can be increased to handle a growing member base without significant architectural changes.

- Conclusion: No insurmountable technical barriers exist. The project can be developed with existing technology and skills.

Financial Feasibility

- Assessment: The proposed system is Financially Feasible with a positive expected Return on Investment (ROI).

- Costs:
  - Development Costs: One-time cost for custom software development. This is the largest upfront investment.
  - Hardware Costs: One-time purchase of digital display, and electronic door locks.
  - Software Costs: Recurring subscriptions for cloud hosting, payment gateway fees, and potential third-party API services.
  - Maintenance & Support: Recurring annual cost for updates, security patches, and technical support (could be ~15-20% of initial development cost per year).

- Benefits & Cost Savings (ROI Justification):

- Reduced Revenue Leakage: Automated subscription validation at the door (FR-02, FR-03) will directly prevent lost income from expired or non-existent memberships. This is a direct financial gain.
- Operational Efficiency: Significant reduction in administrative time spent on manual registration, attendance tracking, and scheduling (Current UC-02, UC-03, UC-04). Staff can be reallocated to higher-value tasks like member service.
- Increased Revenue Streams: Streamlined trainer booking (FR-14) can increase personal training sales. Member retention is likely to improve due to the enhanced experience (progress tracking, convenience), reducing member churn.
- Improved Decision-Making: Data-driven insights from reports (FR-24) can optimize marketing spend, class schedules, and resource allocation, leading to better financial outcomes.
- Conclusion: While there are significant upfront costs, the long-term benefits of increased revenue, reduced losses, and operational savings present a strong business case for a positive ROI, especially when implementing the core modules (BSO 1 & 2) first.

Operational Feasibility

- Assessment: The proposed system is Operationally Feasible and will significantly improve workflows.
- Alignment with Current Procedures: The system is designed to replace inefficient manual processes (paper cards, sign-in books) with streamlined digital ones. It directly addresses the identified pain points.

- Usability and Training:

➢ Members: The interface for scanning QR codes and booking appointments is intuitive and similar to other modern apps. The learning curve is low.

➢ Staff: The admin and instructor dashboards will require training. However, they simplify complex tasks (scheduling, member lookups), so adoption resistance should be low once the benefits are demonstrated.

- Impact on Workflow: The impact is profoundly positive.

- Front Desk: Eliminates manual registration and attendance tracking. Staff become enablers rather than gatekeepers.
- Instructors: Frees them from administrative scheduling tasks, allowing them to focus on coaching.
- Management: Provides real-time visibility into gym operations and member engagement, which is currently non-existent.
- Conclusion: The system will integrate well into the gym's operations. It solves more problems than it creates and is designed with end-user usability in mind.

Organizational Feasibility

- Assessment: The proposed system is Organizationally Feasible, but requires management change.
- Management Support: As outlined in the current system, "Top Management" oversees operations. Their buy-in is critical. The feasibility study and clear ROI presentation are essential to secure their sponsorship.
- User Acceptance: This is the primary potential hurdle.
- Members are likely to welcome the modern convenience, similar to systems used in other industries.
- Staff/Instructors may initially resist due to fear of change or perceived deskilling. This must be managed through:

➢ Clear Communication: Explain the "why" – the system will make their jobs easier, not replace them.

➢ Comprehensive Training: Ensure all staff are confident using the new tools.

➢ Phased Implementation: Start with access control (BSO 1), which has obvious benefits, before rolling out more complex modules.

- Organizational Culture: The shift from a low-tech, manual environment to a data-driven, automated one represents a significant cultural change. Leadership must champion this change to create a culture that embraces efficiency and data.
- Conclusion: The project is viable from an organizational perspective, but its success is contingent on effective change management strategies, strong leadership support, and thorough user training to ensure smooth adoption.

Conclusion

The proposed Gym Management System is feasible across all four evaluated dimensions. It is technically sound, financially justified, operationally beneficial, and organizationally viable with proper management. It is recommended to proceed with the project, adopting a phased implementation approach (e.g., starting with BSO 1: Essential Digitalization & Access Control) to manage risk, budget, and organizational change effectively.

Proposed System Functions Overview

The core functions of the proposed system, as derived from the use cases, are categorized as follows:

A. Member Access & Attendance Management:

- QR-Based Access Control: Members scan a secure QR code at the entrance for identity verification, automated subscription validation, and door unlocking.
- Automated Subscription Validation: A backend service that checks membership status and entitlements in real-time upon access attempts or booking requests.
- Authentication Service: Verifies user identity during the check-in/out process.
- Timestamp Logging: Precisely records all member arrivals and departures, preventing duplicates and ensuring accurate attendance data.
- Staff Access: Manages secure entry and exit for staff members, integrating with the same logging system.

B. Membership & Subscription Management:

- Subscription Purchase & Renewal: Allows members to seamlessly buy or renew subscriptions online via integrated payment gateways, with automatic activation and confirmation.

C. Scheduling & Appointment Management:

- Trainer Appointments: Enables members to view instructor availability, book sessions, and receive automated confirmations and reminders, with checks for subscription eligibility.
- Workout Schedule Modification: Allows members and instructors to edit workout plans, with system checks for conflicts and trainer availability.

D. Member Self-Service & Engagement:

- Workout Completion Tracking: Allows members to mark workouts as complete, triggering progress updates.
- Progress Monitoring & Analytics: Provides members with a dashboard displaying their workout history, trends, consistency metrics, and personal achievements.

E. Administrative & Management Functions:

- Insights Dashboard: Provides instructors and managers with high-level analytics on gym trends, occupancy patterns, and recommended actions.

- Detailed Reporting: Allows authorized managers to generate and export custom reports on attendance, revenue, and facility utilization.
- System Administration: A comprehensive console for administrators to manage all system aspects, including user roles, pricing plans, facility configurations, and hardware settings.

Diagrams

Use-Case Diagram

This section presents the overall use case diagram for the proposed Gym Management System, illustrating the main interactions between the key actors - members, trainers, managers, and administrators. The diagram highlights the primary activities available to each actor within the integrated platform and provides a visual summary of the system's intended functionality and scope.

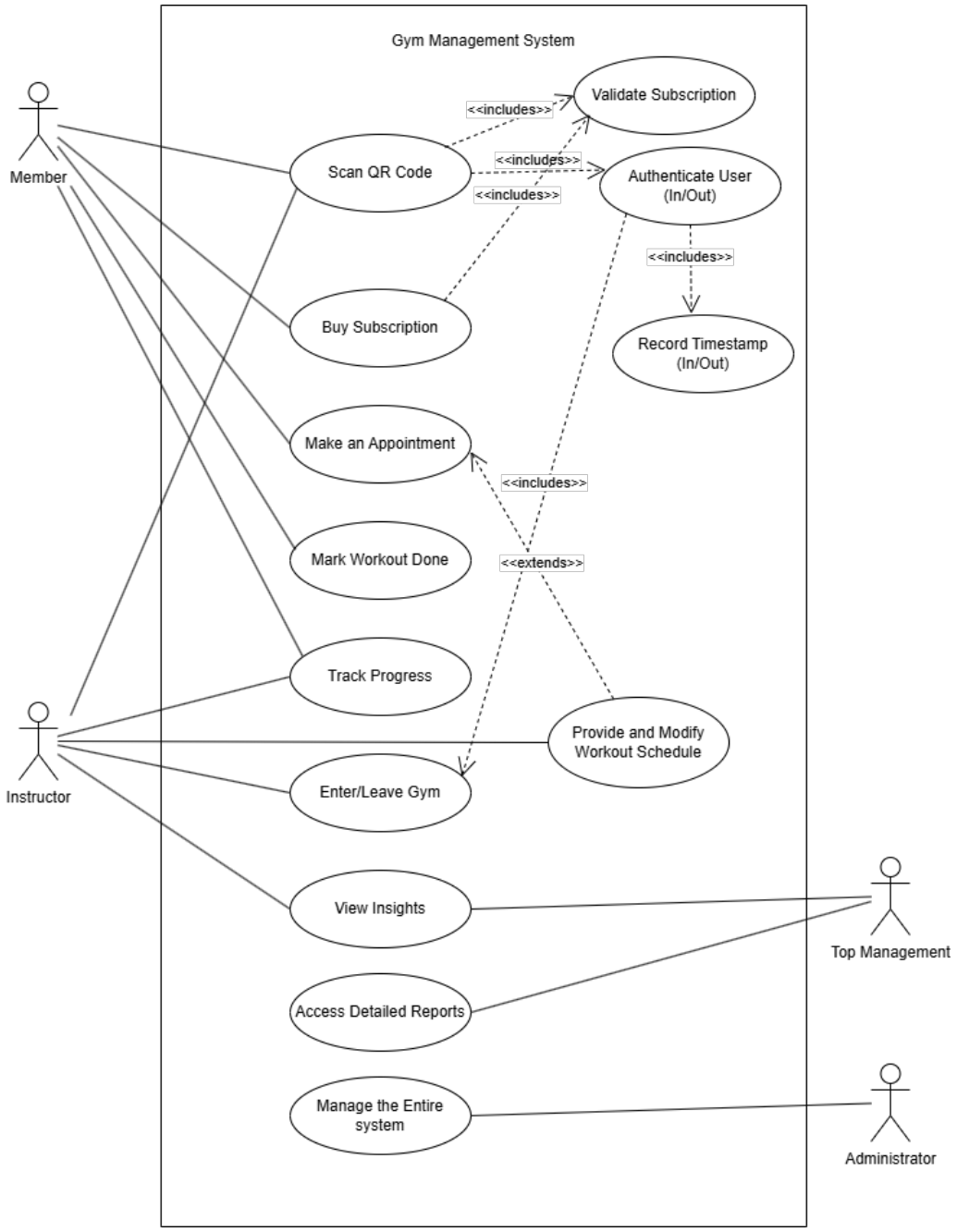


Figure 1-2: Use Case Diagram for Proposed System

Use-Case Descriptors

UC-01 — Scan QR Code

Field	**Description**
Brief	Member scans a gym QR code (at the entrance) so the system can identify them, validate membership and log their arrival/departure.
Primary actor	Member (or Staff scanning on member's behalf)
Preconditions	Member has a valid account in the system; QR code is displayed using a display; network available.
Postconditions (success)	Member is authenticated (door opened), subscription validated, and entrance (or exit) timestamp is recorded.
Postconditions (failure)	Scan fails or authentication/validation fails; no timestamp recorded; member is shown an error and instructions.
Trigger	Member (or staff) presents QR; device scans it.
Main success scenario	1. Member opens scanner or presents printed QR to kiosk. 2. Scanner reads QR code (memberID / session token / check-in code). 3. System fetches member profile using QR data. 4. System calls Validate subscription (UC-02). 5. System calls Authenticate user (UC-03) to confirm identity/session. 6. System calls to unlock the door. 7. System calls Record timestamp (UC-04) to log arrival/exit.

Extensions / alternate flows	QR unreadable → scanner asks to retry or type member ID. Subscription invalid/expired → prompt to Buy subscription (UC-05) or display restricted access message. Authentication fails → display the error message. Timestamp logging unavailable → store locally.

UC-02 — Validate subscription

Field	**Description**
Brief	Check membership/subscription status and entitlements for a member.
Primary actor	Gym System (triggered by other use cases)
Preconditions	Member record exists; subscription data available.
Postconditions (success)	Returns valid/invalid status and entitlements (access hours, facility restrictions).
Postconditions (failure)	Unable to determine status (service down) or subscription not valid.
Trigger	Called by scanning, entry attempts, or feature access requests (e.g., booking class).
Main success scenario	1. Request arrives with member ID. 2. Service looks up active subscriptions and entitlements. 3. Service returns validation result and expiry/limits.

Extensions / alternate flows	Member has multiple subscriptions — resolve priority and return effective entitlement. Any failure → return error code and message prompting user action (renew, contact support).

UC-03 — Authenticate user (in/out)

Field	**Description**
Brief	Verify the identity of the person scanning in/out and opening the door.
Primary actor	Member; Staff (for manual auth)
Preconditions	Member has authentication credentials.
Postconditions (success)	Member is marked authenticated, door unlocked.
Postconditions (failure)	Authentication denied; entry blocked.
Trigger	Called by Scan QR Code.
Main success scenario	1. Member scans the QR Code and door unlocks.
Extensions / alternate flows	1. Invalid QR Code. 2. Subscription Expired. 3. QR Code Timed out.

UC-04 — Record timestamp (in/out)

Field	**Description**
Brief	Create a log entry for member arrival or departure.

Primary actor	Gym System (backend), triggered by scan/auth events
Preconditions	Member authenticated and subscription validated (if required).
Postconditions (success)	Persistent timestamp record with memberID, time, location and event type (IN/OUT).
Postconditions (failure)	No record created.
Trigger	Authentication success; manual check-in/out by staff; scheduled auto-checkout.
Main success scenario	1. Received event (memberID, IN/OUT). 2. Validate event ordering (prevent duplicate INs or OUTs). 3. Persist timestamp with metadata (deviceID, gateID). 4. Return acknowledgement.
Extensions / alternate flows	Duplicate IN detected → flag for staff review and optionally deny second IN. DB write fails → notify the administrator.

UC-05 — Buy subscription

Field	**Description**
Brief	Member purchases or renews a subscription via the app or website.
Primary actor	Member
Preconditions	Member account exists; payment method available.
Postconditions (success)	Subscription becomes active or extended and member receives confirmation; subscription data updated in system.
Postconditions (failure)	Payment fails; system rolls back partial changes; member notified.
Trigger	Member selects new plan or renewal from UI.
Main success scenario	1. Member selects plan and payment method. 2. System shows price & terms; member confirms. 3. System processes payment via gateway.

	4. Billing service activates subscription and updates member record. 5. System calls Validate subscription to ensure new status is visible. 6. Confirmation sent to member.
Extensions / alternate flows	Payment declined → show error, allow retry or alternate payment. Activation fails → scheduled activation and notify member with expected time.

UC-06 — Make an appointment

Field	**Description**
Brief	Member schedules a trainer appointment.
Primary actor	Member, Instructor
Preconditions	Member authenticated; scheduling data (availability) up-to-date.
Postconditions (success)	Appointment is created, calendar updated, notifications sent to member/trainer.
Postconditions (failure)	Time slot unavailable; appointment not created.
Trigger	Member selects time slot and preferred instructor in app.
Main success scenario	1. Member browses available slots, instructor and selects one. 2. System verifies member entitlement via Validate subscription. 3. System confirms slot availability and reserves it. 4. System saves appointment and notifies participants. 5. Optionally, system may create a calendar entry and reminder.

Extensions / alternate flows	Insufficient entitlement → prompt to upgrade subscription (UC-05). Slot was taken concurrently → present next available alternatives. Member cancels → apply cancellation policy and notify trainer.

UC-07 — Mark workout done

Field	**Description**
Brief	Member marks a scheduled workout or completed routine as done (for progress tracking).
Primary actor	Member
Preconditions	Member has a workout scheduled or a routine to complete.
Postconditions (success)	Workout status marked complete, progress metrics updated, timestamp recorded.
Postconditions (failure)	System fails to update progress; member warned and retry option given.
Trigger	Member presses “Done” in app, or system auto-detects completion (optional) – upon leaving.
Main success scenario	1. Member opens the session and marks it completed. 2. System verifies session identity and marks status complete. 3. System calls Record timestamp to log completion time. 4. Progress statistics are recalculated and updates shown to member.
Extensions / alternate flows	Member attempts to mark before required activities finished → prompt confirmation. Offline → persist locally and sync later.

UC-08 — View / Track progress

Field	Description
Brief	Member views historical workout data, trends and statistics.
Primary actor	Member
Preconditions	Member has historical workout data (timestamps, routines, measurements).
Postconditions (success)	Dashboard is displayed with up-to-date metrics and visualizations.
Postconditions (failure)	Partial data shown with message, or no data if none exists.
Trigger	Member opens progress page.
Main success scenario	1. Member navigates to progress screen. 2. System gathers activity history, workout completion, and measurement data. 3. Analytics engine computes trends (consistency, volume, personal records). 4. System displays graphs, summaries and suggestions.
Extensions / alternate flows	System permissions missing (e.g., wearable integration) → show authorized steps. Insufficient data → show tips to get started.

UC-09 — Enter / Leave Gym

Field	Description
Brief	Use case representing physical access for staff; may extend or be realized by scanning QR, staff presence within gym.
Primary actor	Staff
Preconditions	Door hardware functioning; Gym Wi-Fi connection is up and running.
Postconditions (success)	Staff member enters or leaves; access record logged.
Postconditions (failure)	Entry denied and staff alerted.
Trigger	Staff member attempts to enter or exit facility.
Main success scenario	1. Member requests access (QR scan) or presence detected via Wi-Fi.

	2. System validates and authenticates (UC-02, UC-03). 3. Door unlocks, and Record timestamp captures IN/OUT.

UC-10 — Change / Modify workout schedule

Field	**Description**
Brief	Member or instructor edits a workout plan.
Primary actor	Member; Instructor
Preconditions	User authenticated and authorized to change the schedule.
Postconditions (success)	Schedule updated; affected participants notified.
Postconditions (failure)	Conflict detected and change rejected or queued for approval.
Trigger	User selects edit/reschedule.
Main success scenario	1. User opens schedule and edits exercises/times. 2. System verifies constraints (trainer availability, subscription entitlements). 3. System saves changes and sends notifications.
Extensions / alternate flows	Conflicting appointments → present alternatives or request approval.

UC-11 — View insights

Field	**Description**
Brief	Instructor or manager views higher-level insights (trends, recommended actions, occupancy patterns).
Primary actor	Manager; Instructor
Preconditions	Sufficient aggregated data available.
Postconditions (success)	Insights generated and displayed; optionally action items suggested.
Postconditions (failure)	Insufficient data for certain insights → system limits results and suggests how to collect more.

Trigger	User opens Insights page or system sends proactive insights.
Main success scenario	1. User requests insights. 2. System runs analytics on activity, attendance, and performance. 3. Results and suggestions displayed.

UC-12 — Access detailed reports

Field	**Description**
Brief	Managers or authorized staff generate/export detailed reports (attendance, revenue, utilization).
Primary actor	Manager / Admin
Preconditions	User has proper role and permissions.
Postconditions (success)	Report generated and downloadable; logs of access stored.
Postconditions (failure)	Report generation error; user informed.
Trigger	Manager selects report parameters and requests generation.
Main success scenario	1. Manager picks report type, date range and filters. 2. System validates permission and retrieves data. 3. Report compiled and presented; export options provided.

UC-13 — Manage the entire system

Field	**Description**
Brief	Admin-level operations: configure facilities, user roles, pricing plans, hardware endpoints, and overall system settings.
Primary actor	System Administrator
Preconditions	Admin authenticated with appropriate privileges.
Postconditions (success)	System configuration updated; changes applied to dependent services.

Postconditions (failure)	Partial updates rolled back; admin alerted.
Trigger	Admin chooses management task (create plan, change pricing, add location).
Main success scenario	1. Admin opens Admin console. 2. Admin selects a management function (e.g., create a subscription plan). 3. System validates inputs and saves changes. 4. System logs action and informs affected components/users if necessary.

Activity Diagrams

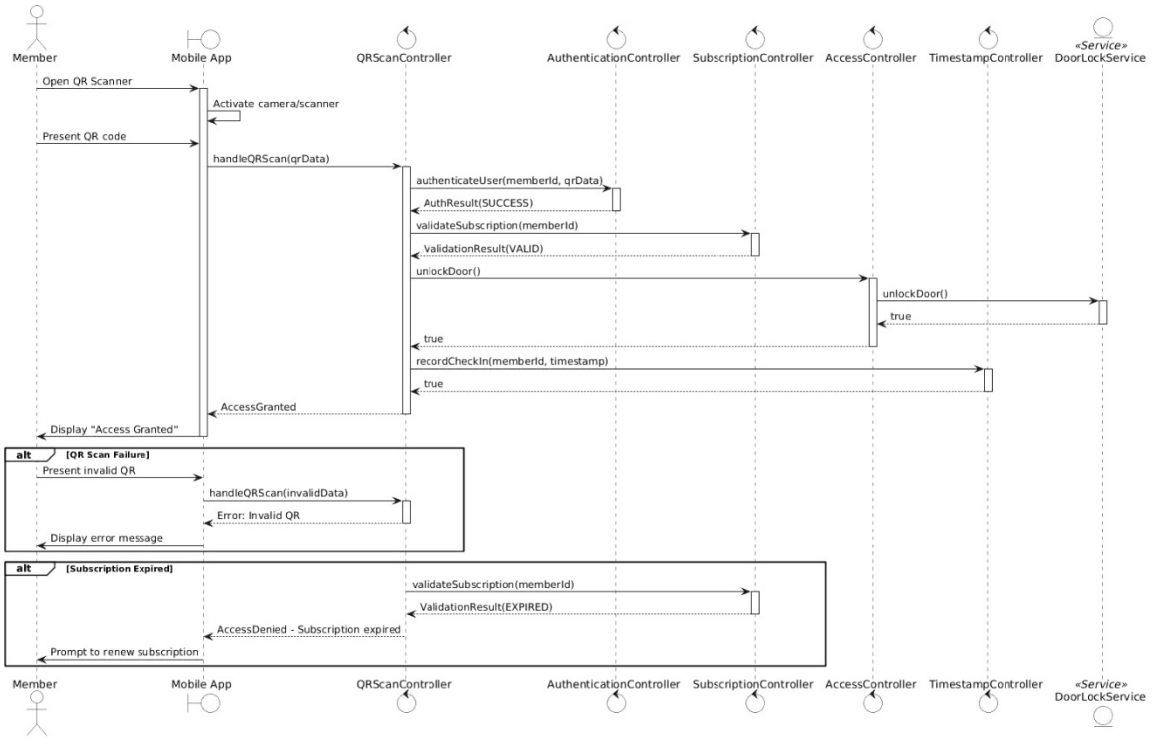
Activity diagrams are used here to visually represent the workflows of the **proposed system**. Each diagram outlines the sequence of actions, decisions, and interactions between system components and users, showing how tasks will progress from one stage to another in the new design. By mapping these processes, the diagrams provide a clear understanding of how activities will be automated, coordinated, and managed within the proposed system, highlighting the improved flow of operations and decision-making.

Class Diagrams

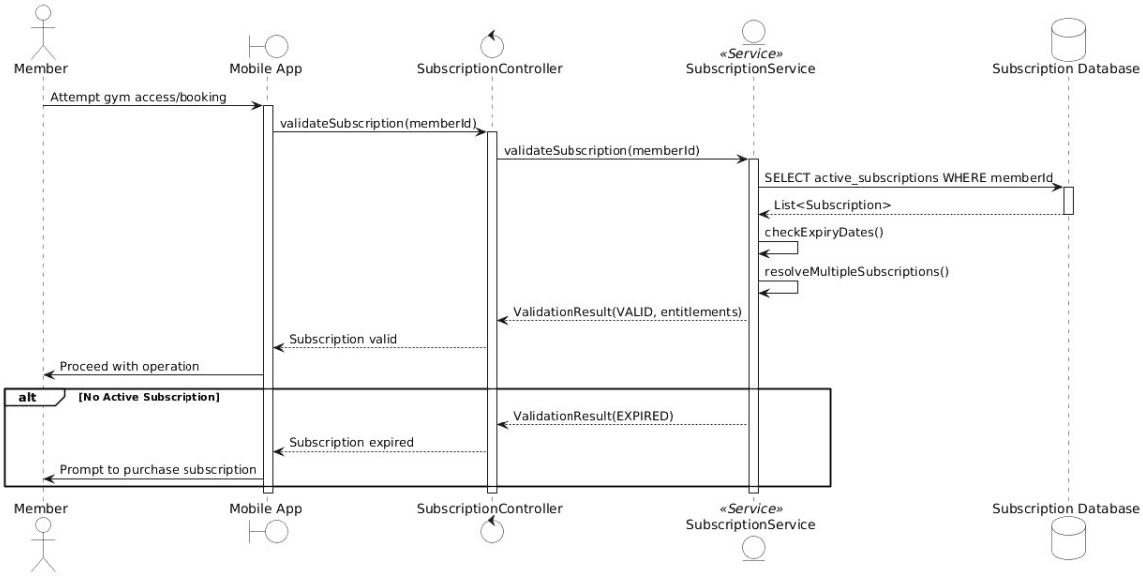
Sequence Diagrams

Sequence diagrams describe how various system elements interact in steps for specific processes, such as an order being placed or inventory updates. They capture the sequence of messages between users and the system and are hence indicative of process flow.

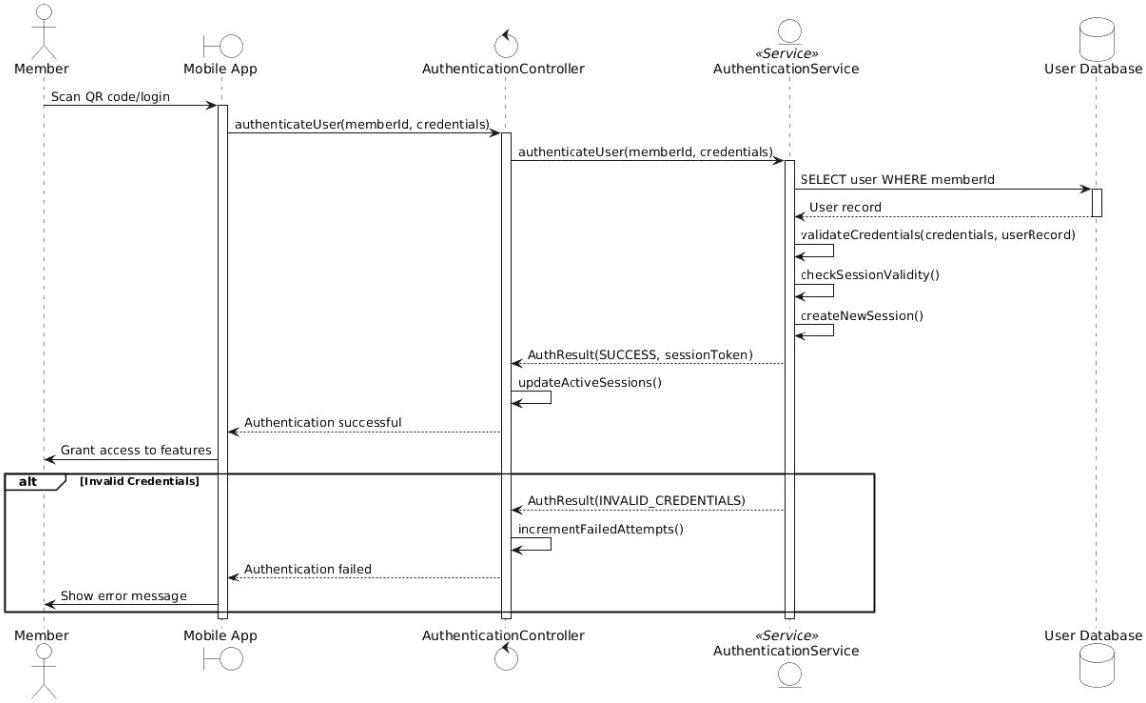
UC-01: Scan QR Code - Sequence Diagram



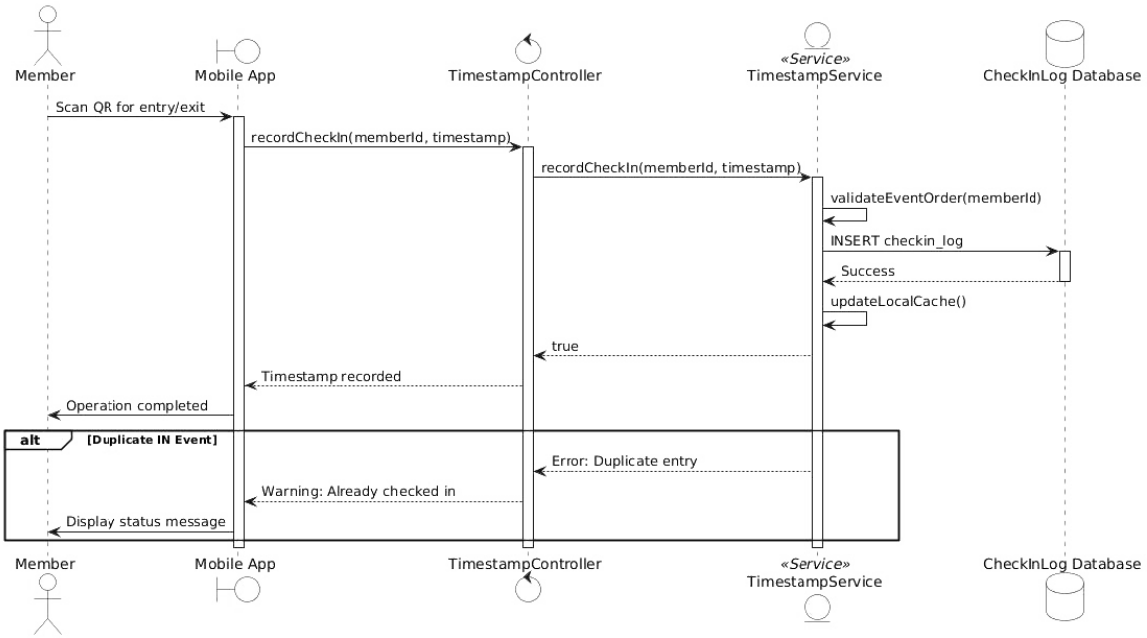
UC-02: Validate Subscription - Sequence Diagram



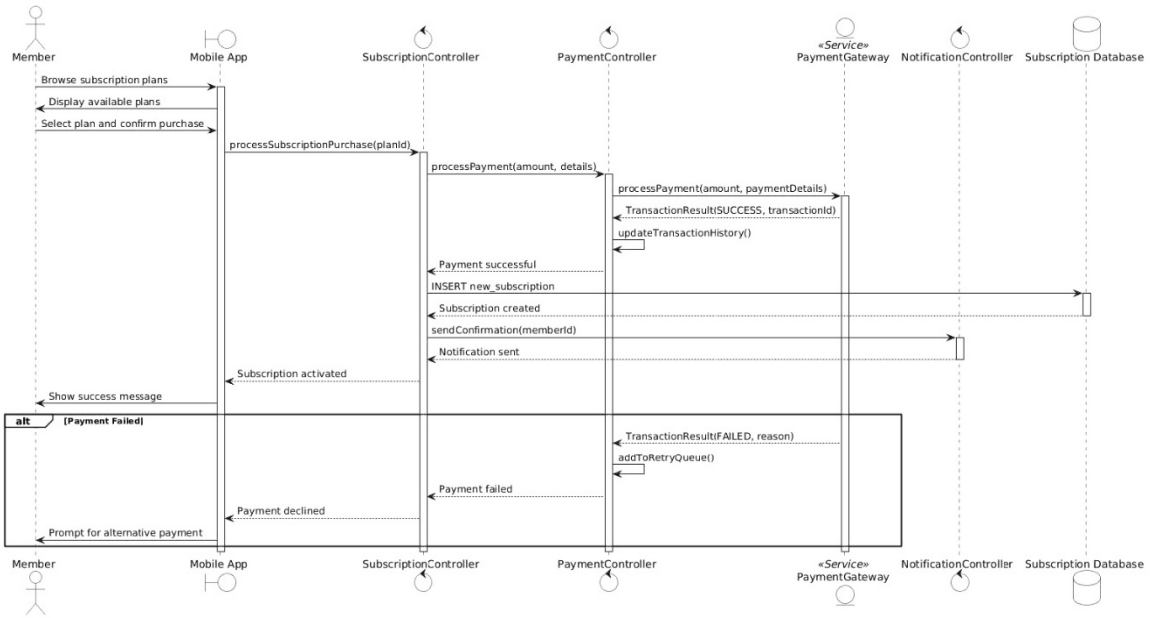
UC-03: Authenticate User - Sequence Diagram



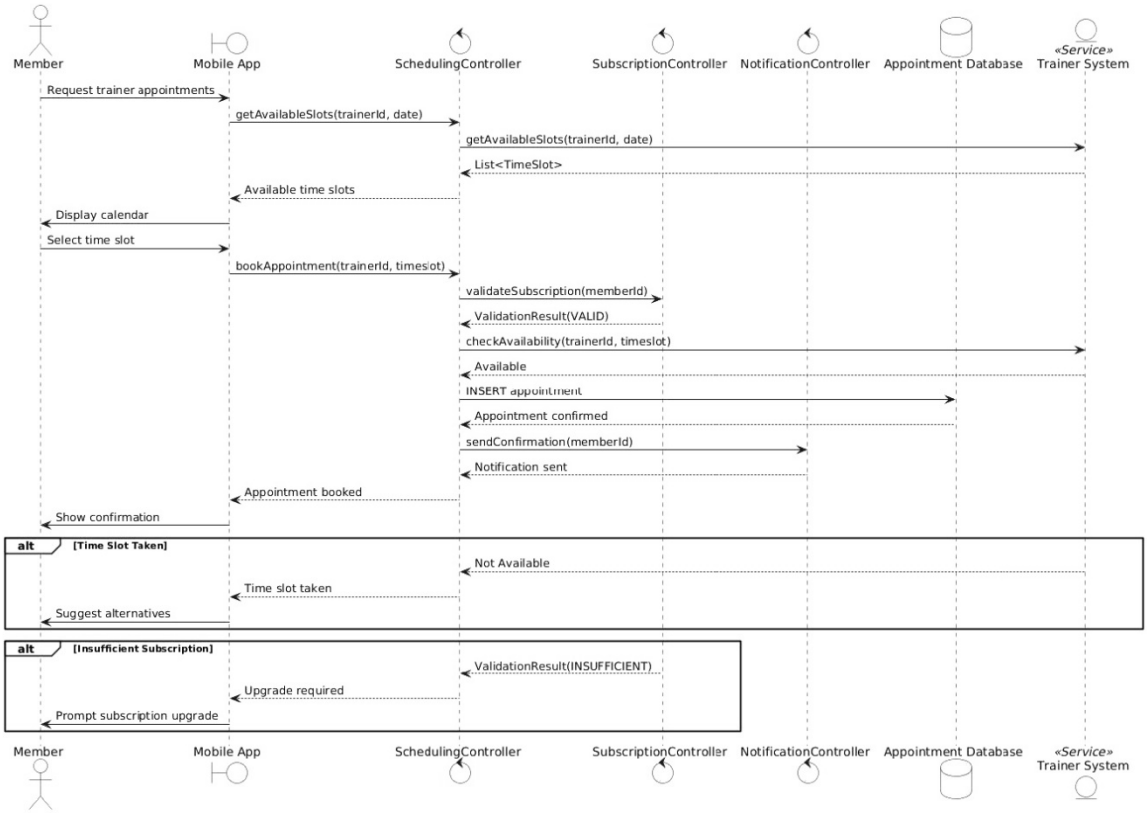
UC-04: Record Timestamp - Sequence Diagram



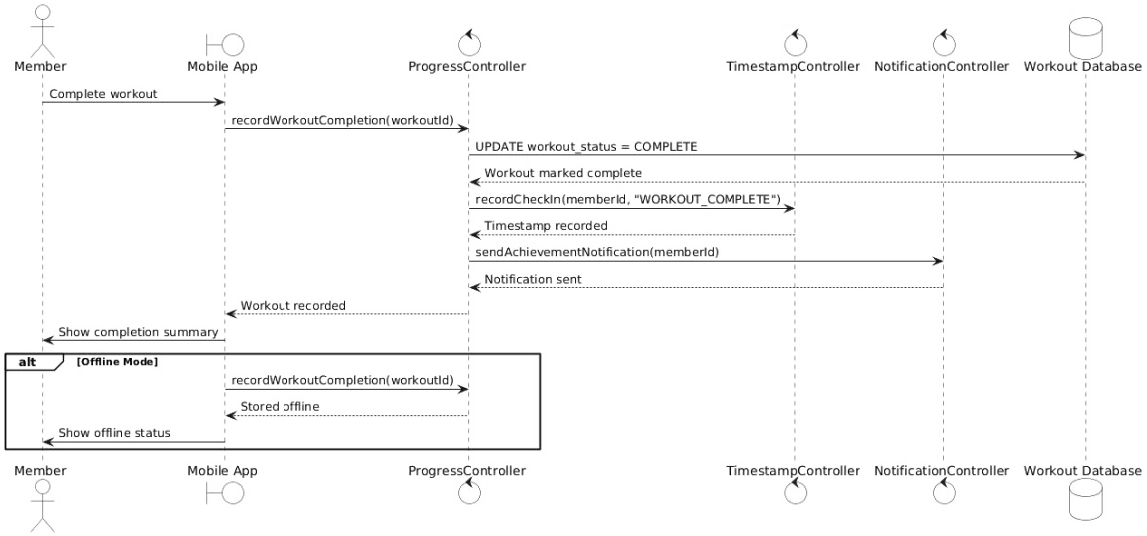
UC-05: Buy Subscription - Sequence Diagram



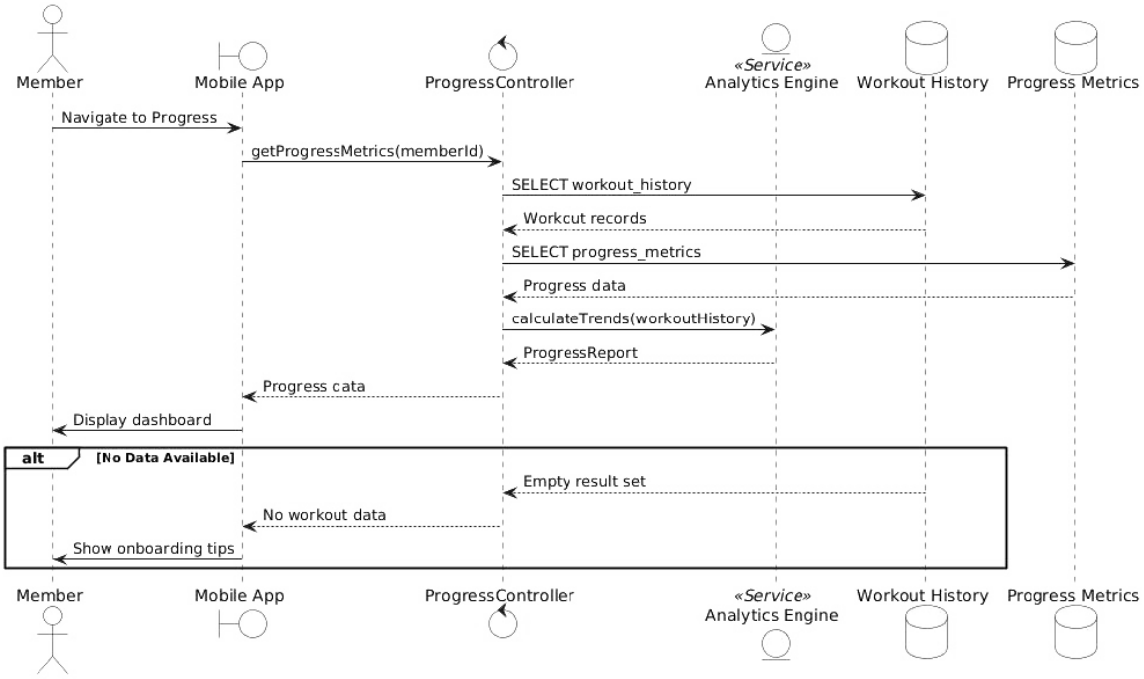
UC-06: Make an Appointment - Sequence Diagram



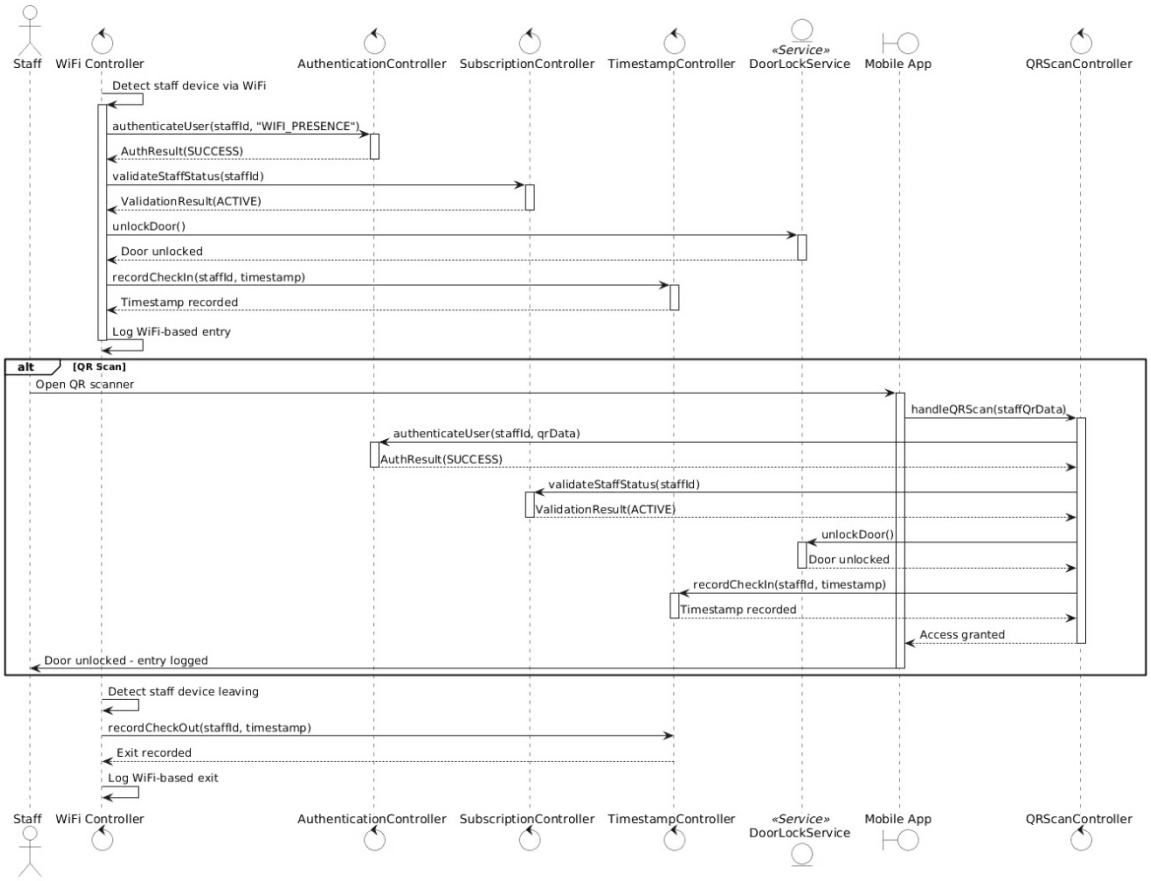
UC-07: Mark Workout Done - Sequence Diagram



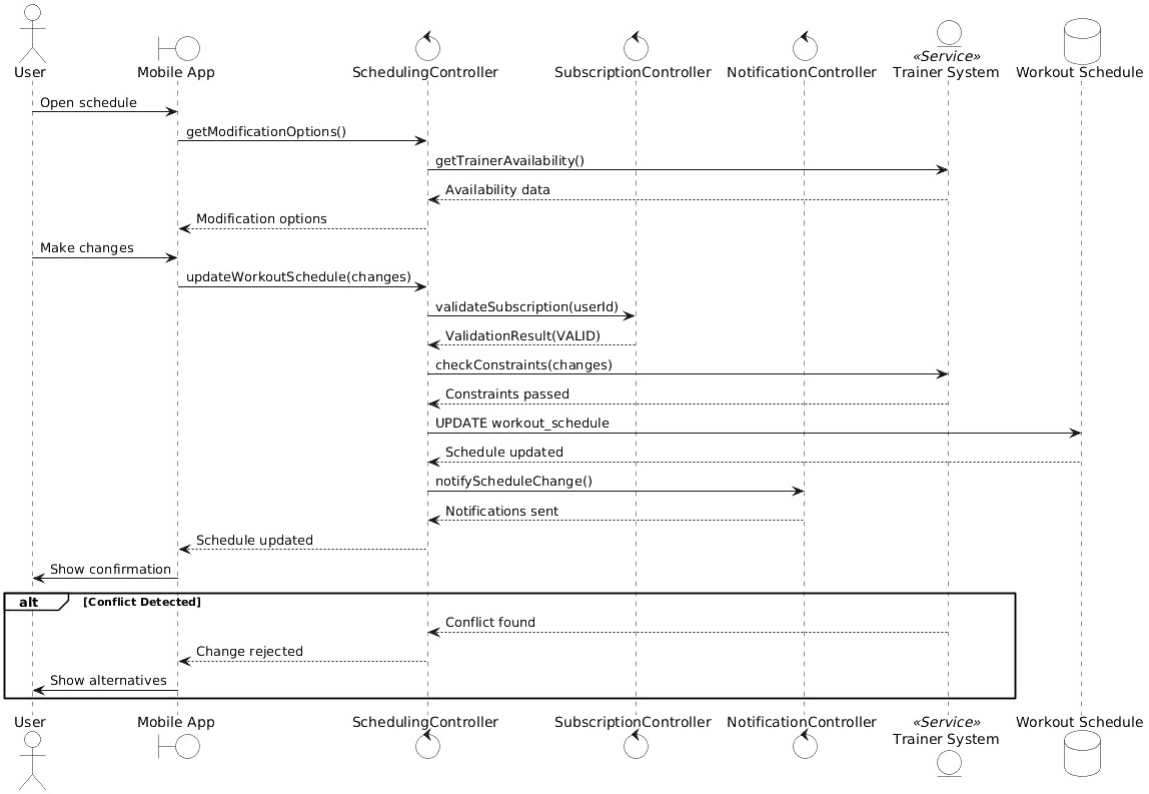
UC-08: View/Track Progress - Sequence Diagram



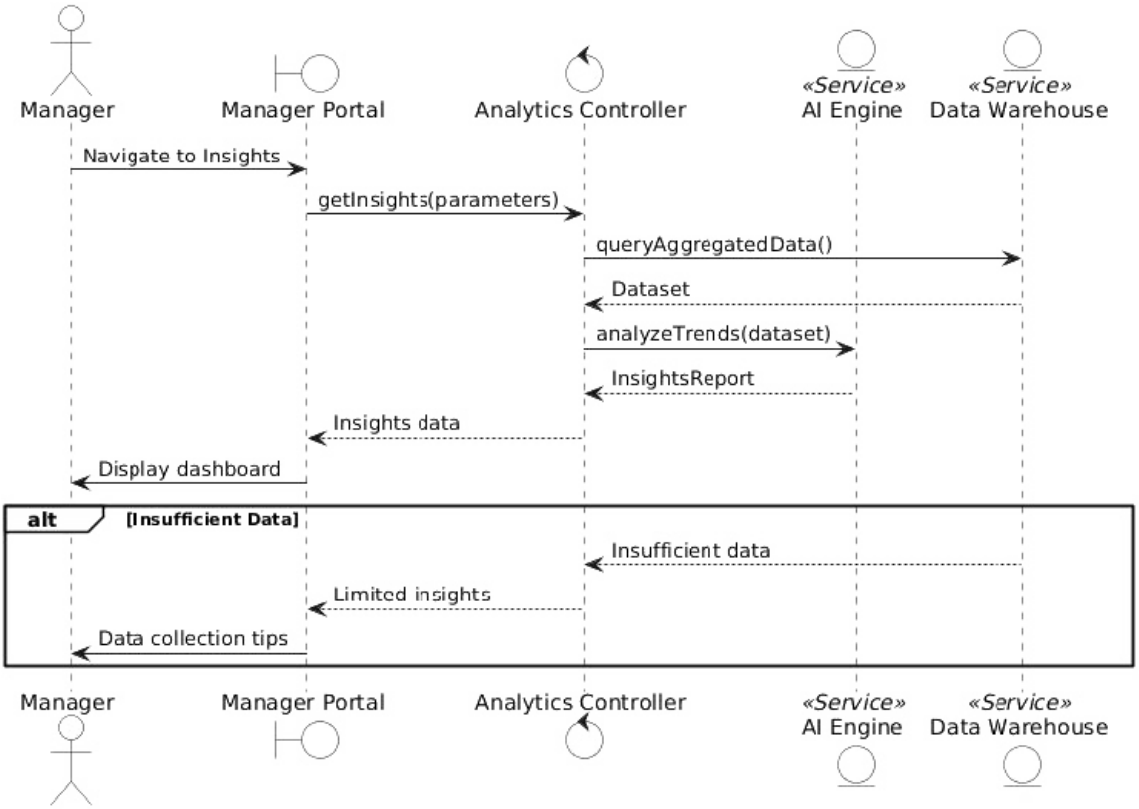
UC-09: Enter/Leave Gym (Staff) - Sequence Diagram



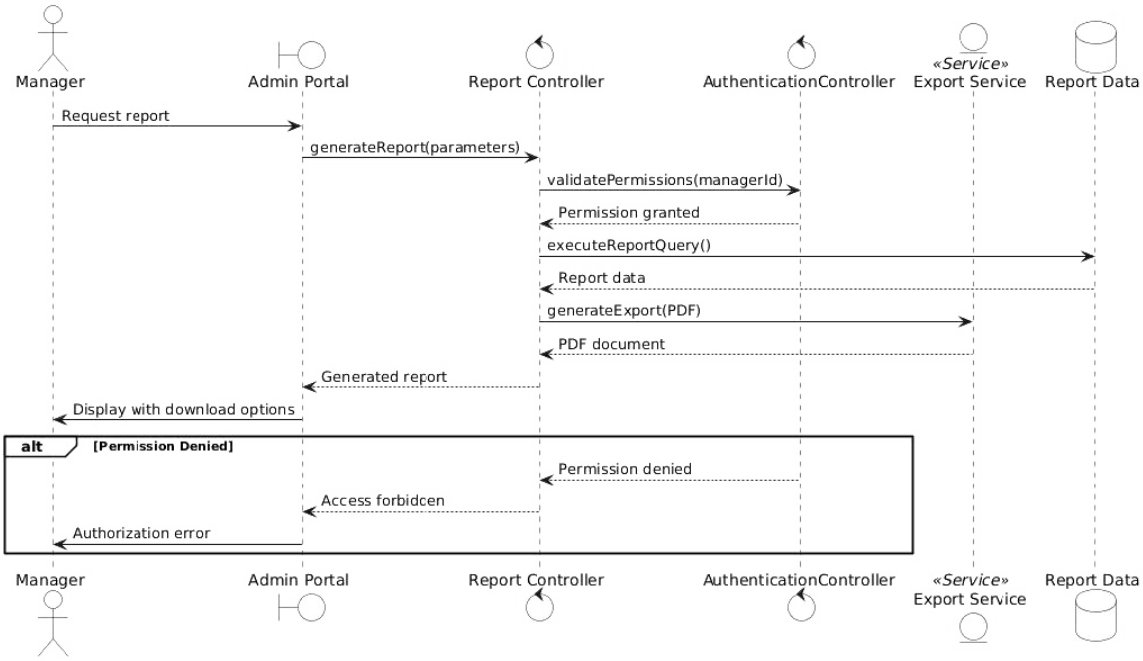
UC-10: Change/Modify Workout Schedule - Sequence Diagram



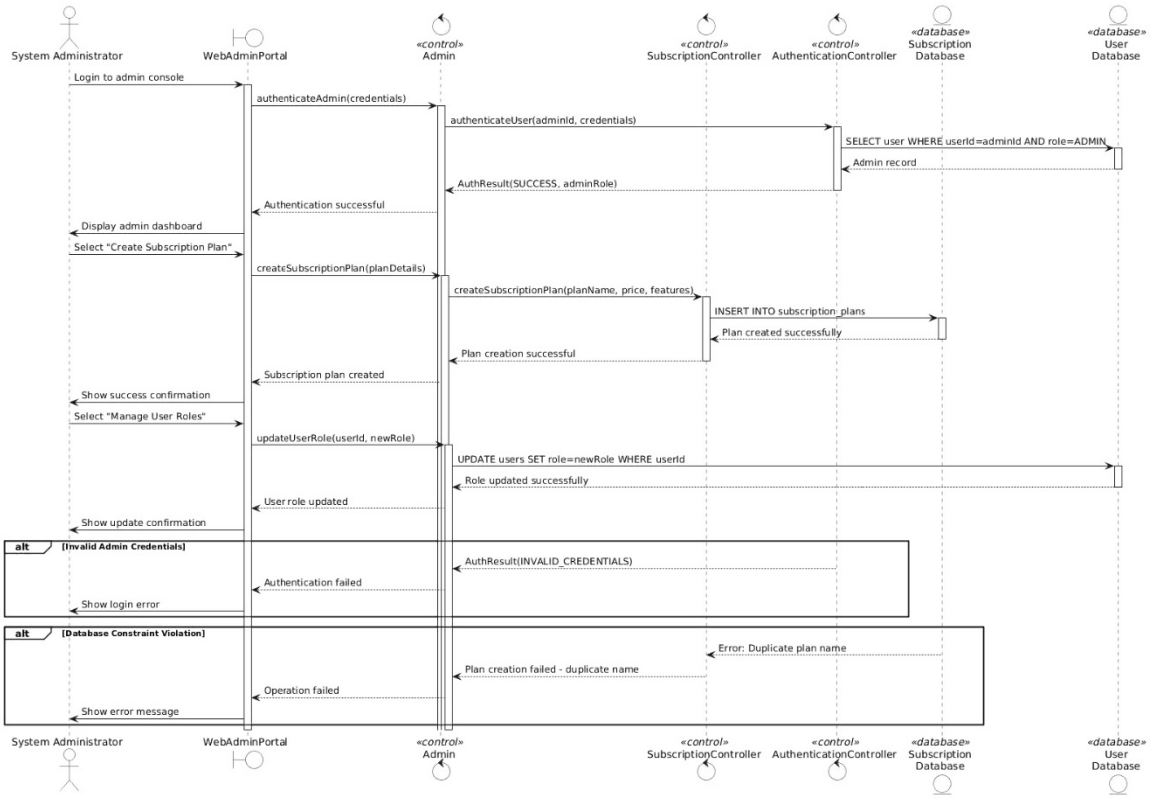
UC-11: View Insights - Sequence Diagram



UC-12: Access Detailed Reports - Sequence Diagram



UC-13: Manage Entire System - Sequence Diagram



Graphical User Interfaces

The User Interface (UI) of the Gym Management System is designed to provide a simple and user-friendly experience. It focuses on clarity, ease of use, and smooth navigation to help users perform tasks efficiently. The UI is developed using HTML, ensuring a well-structured and visually consistent layout.

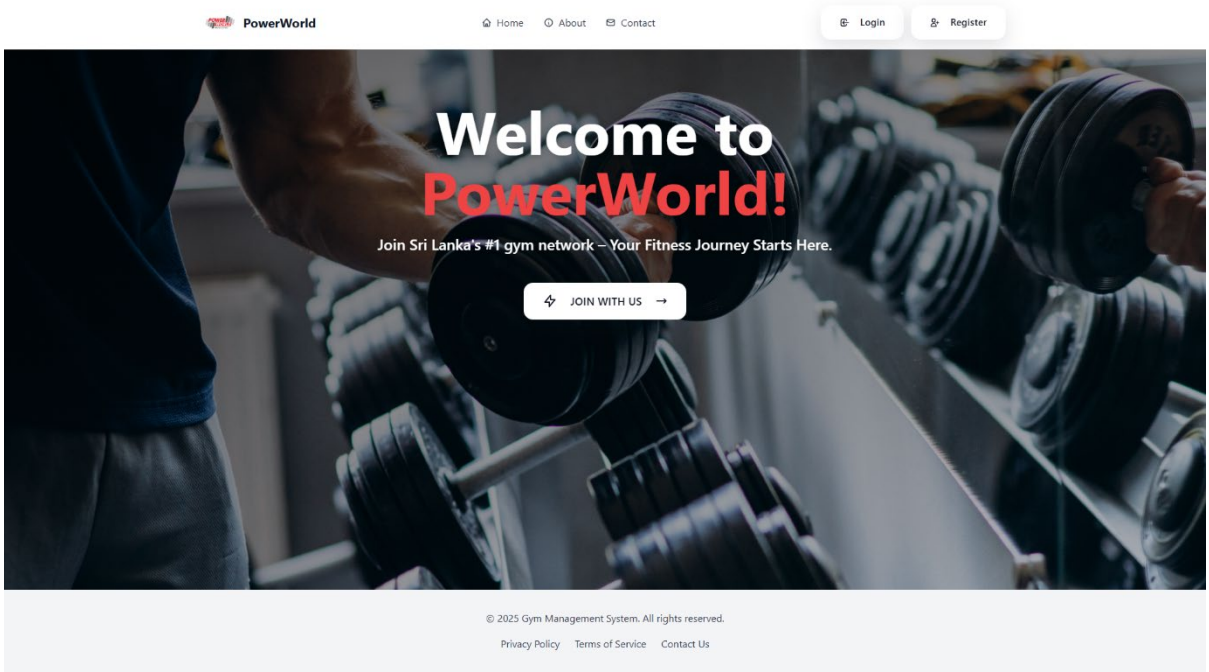


Figure 1: Homepage

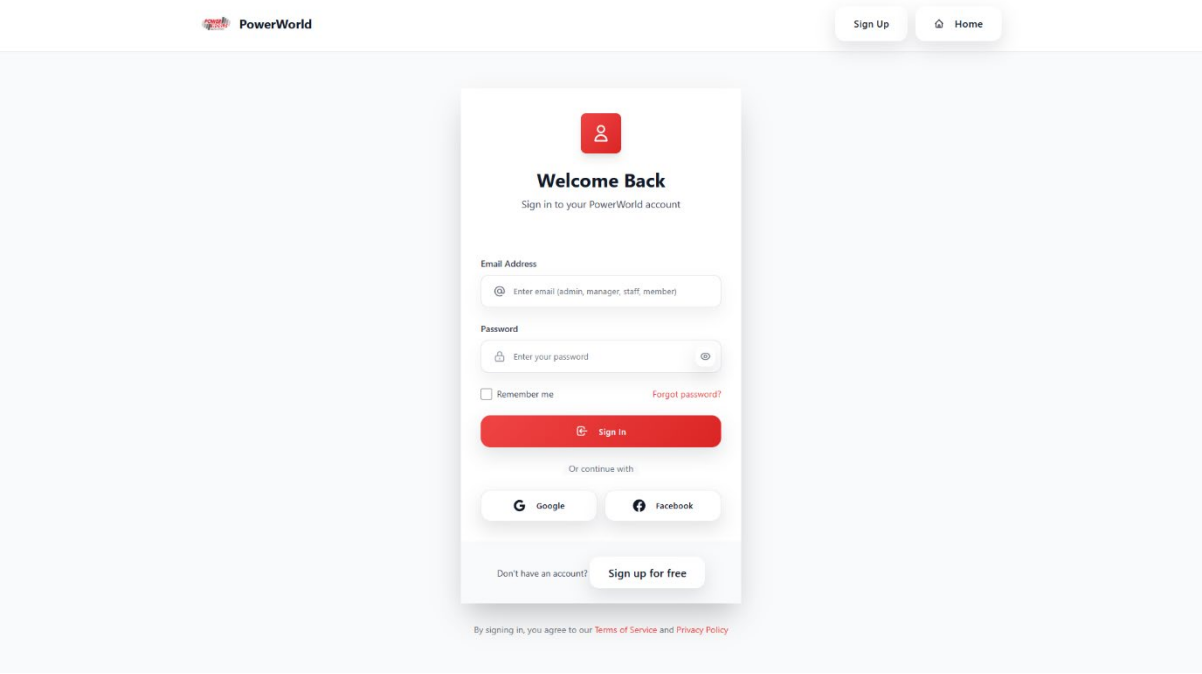


Figure 2: Sign-in Screen

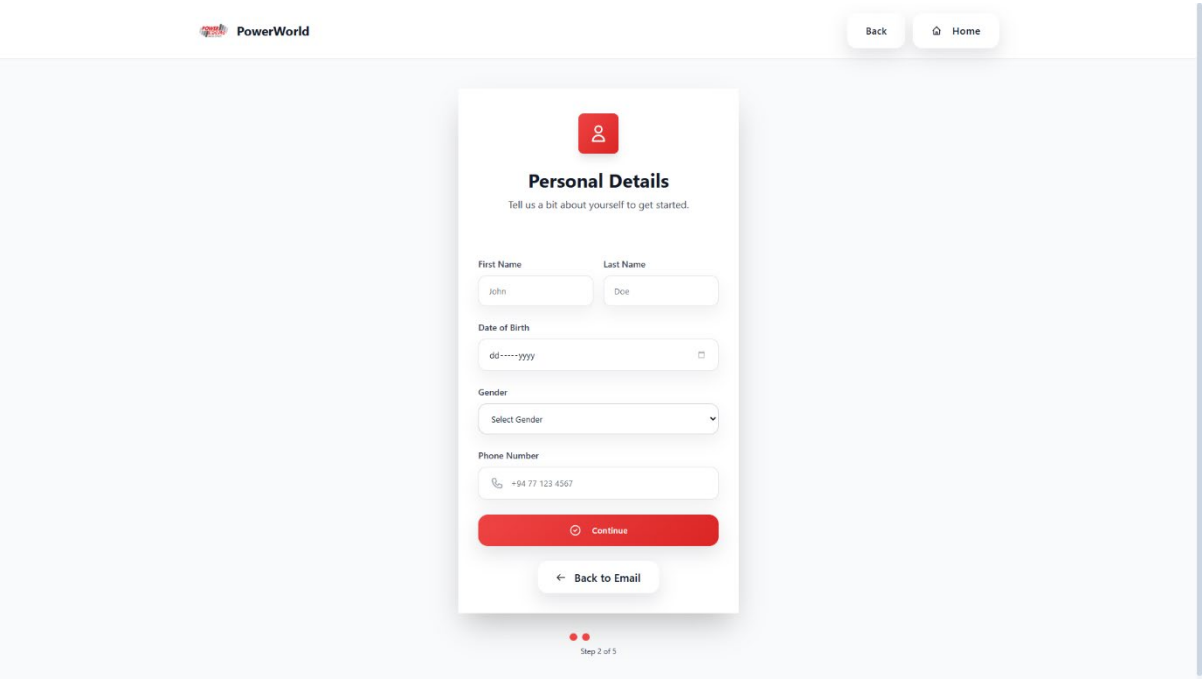


Figure 3: Sign-Up asking for personal info

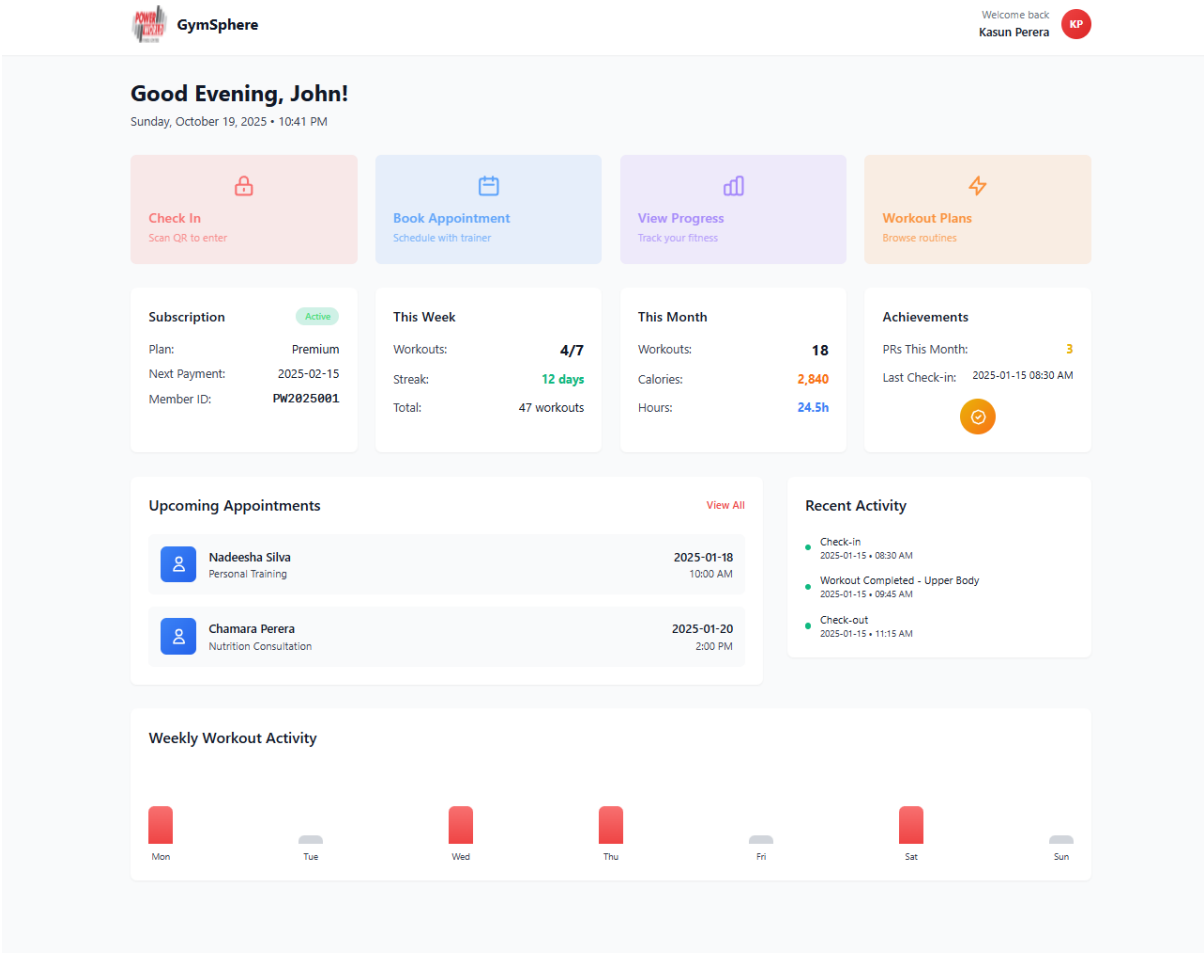


Figure 4: Member Dashboard

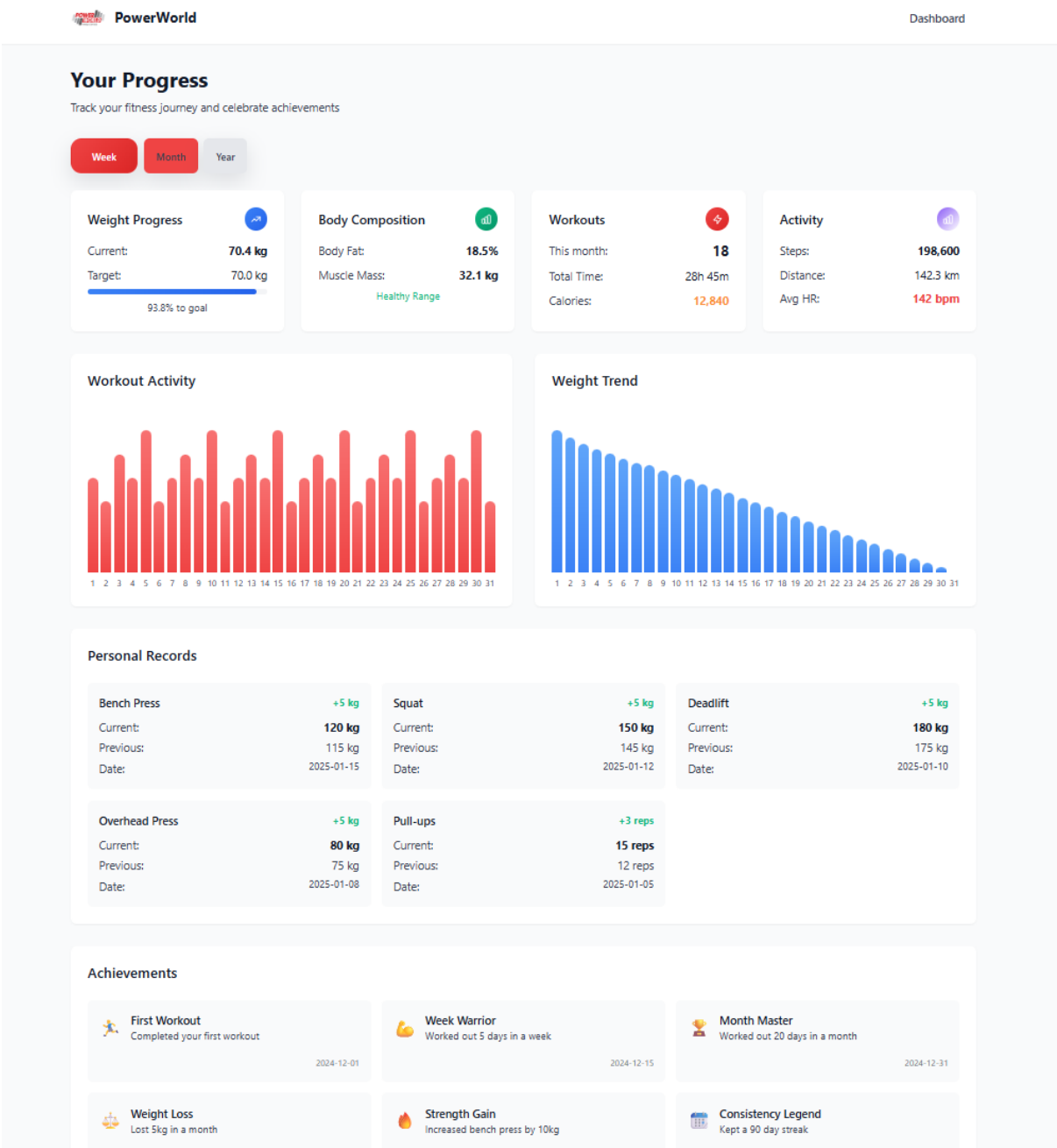


Figure 5: Member Progress Tracking

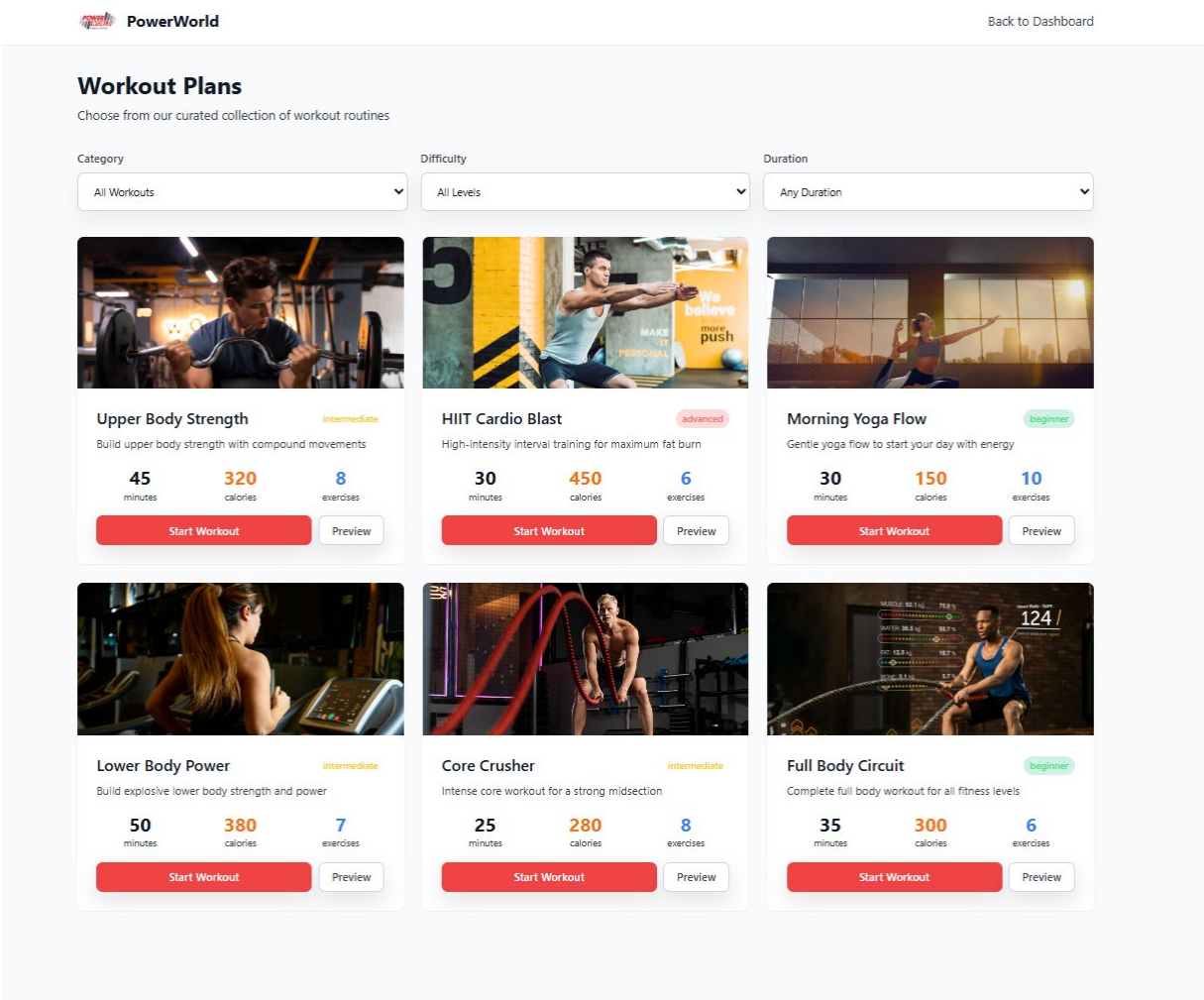


Figure 6: Workout Plans

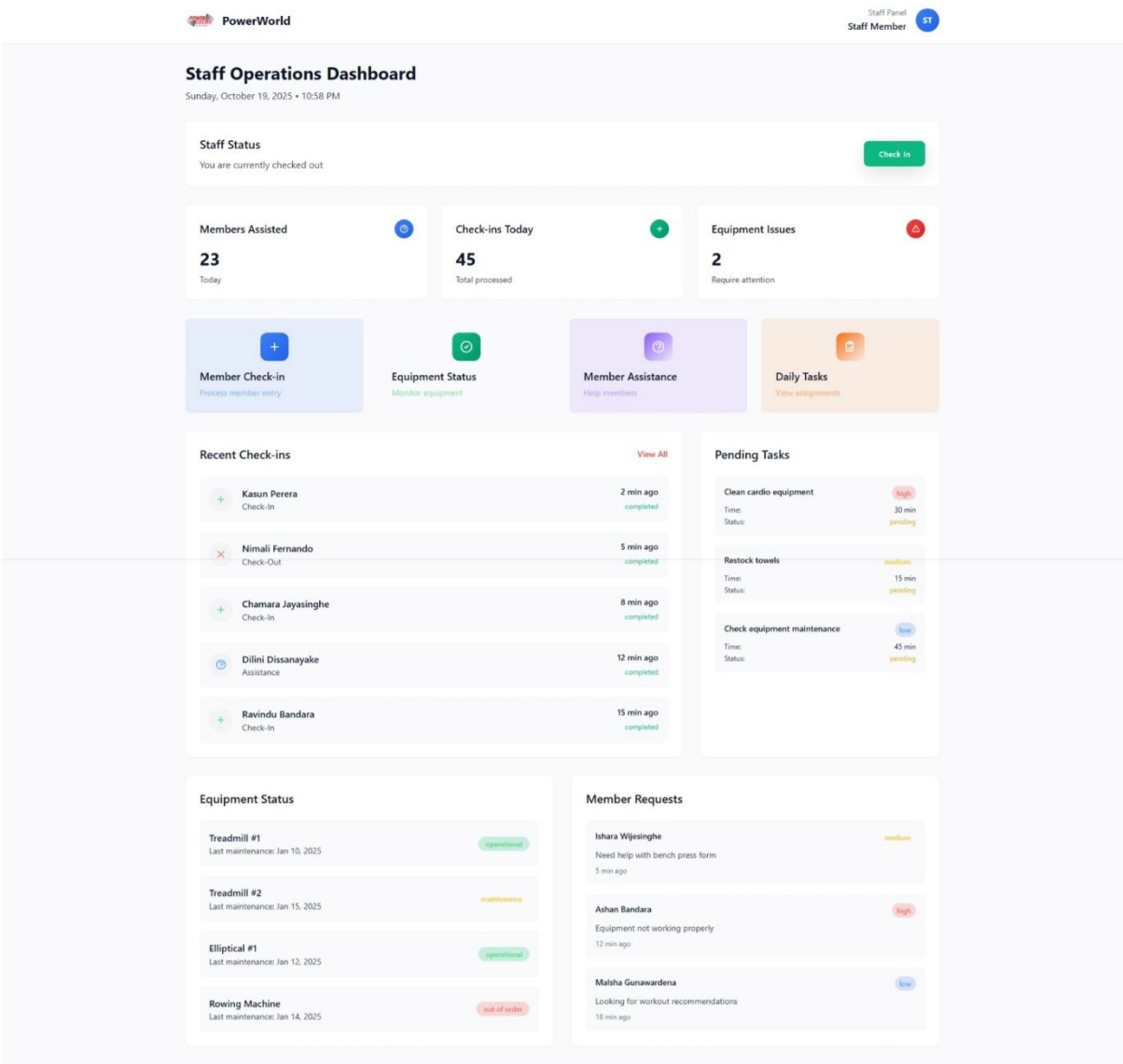


Figure 7: Staff Dashboard

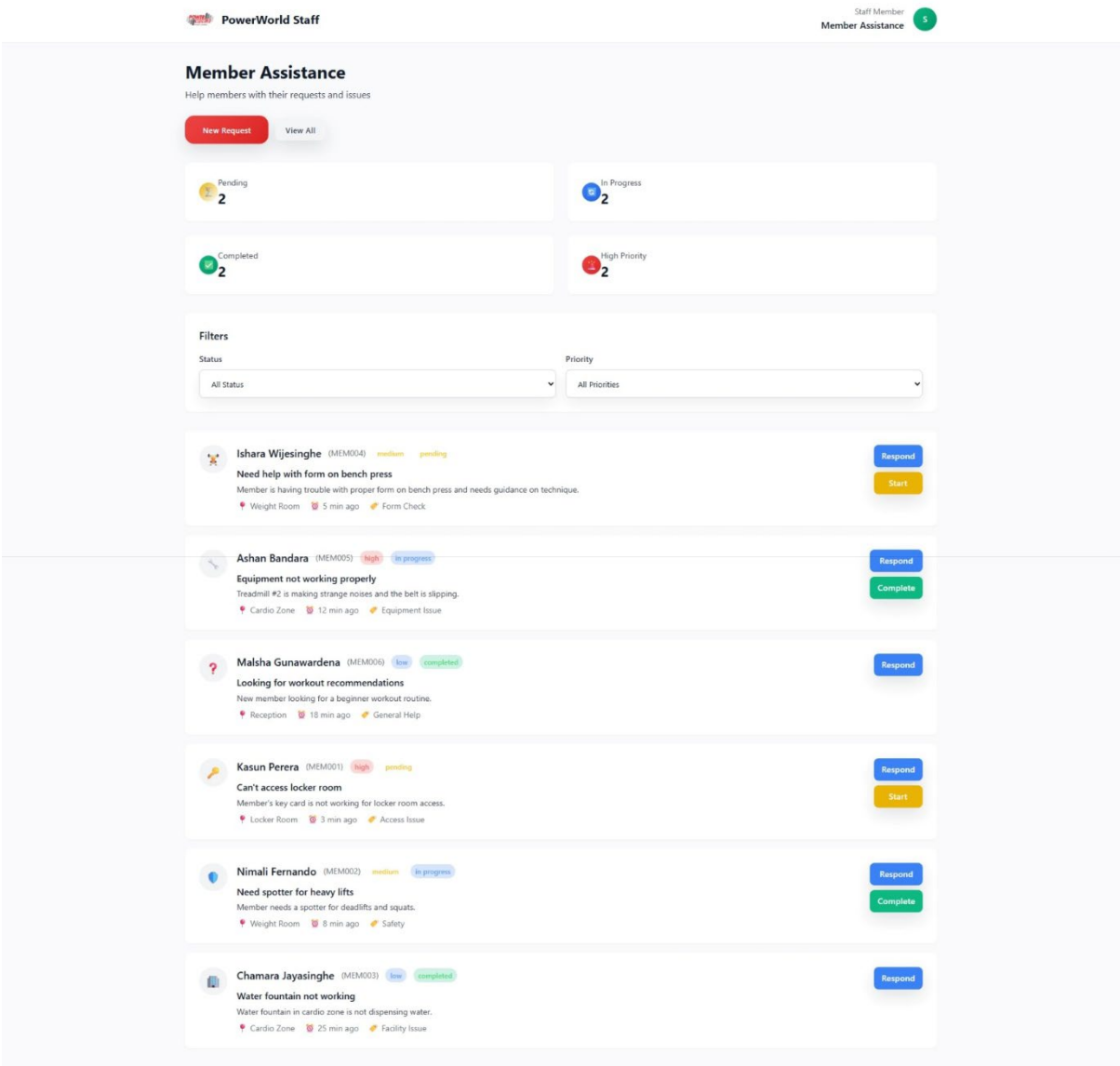


Figure 8: Member Assistance

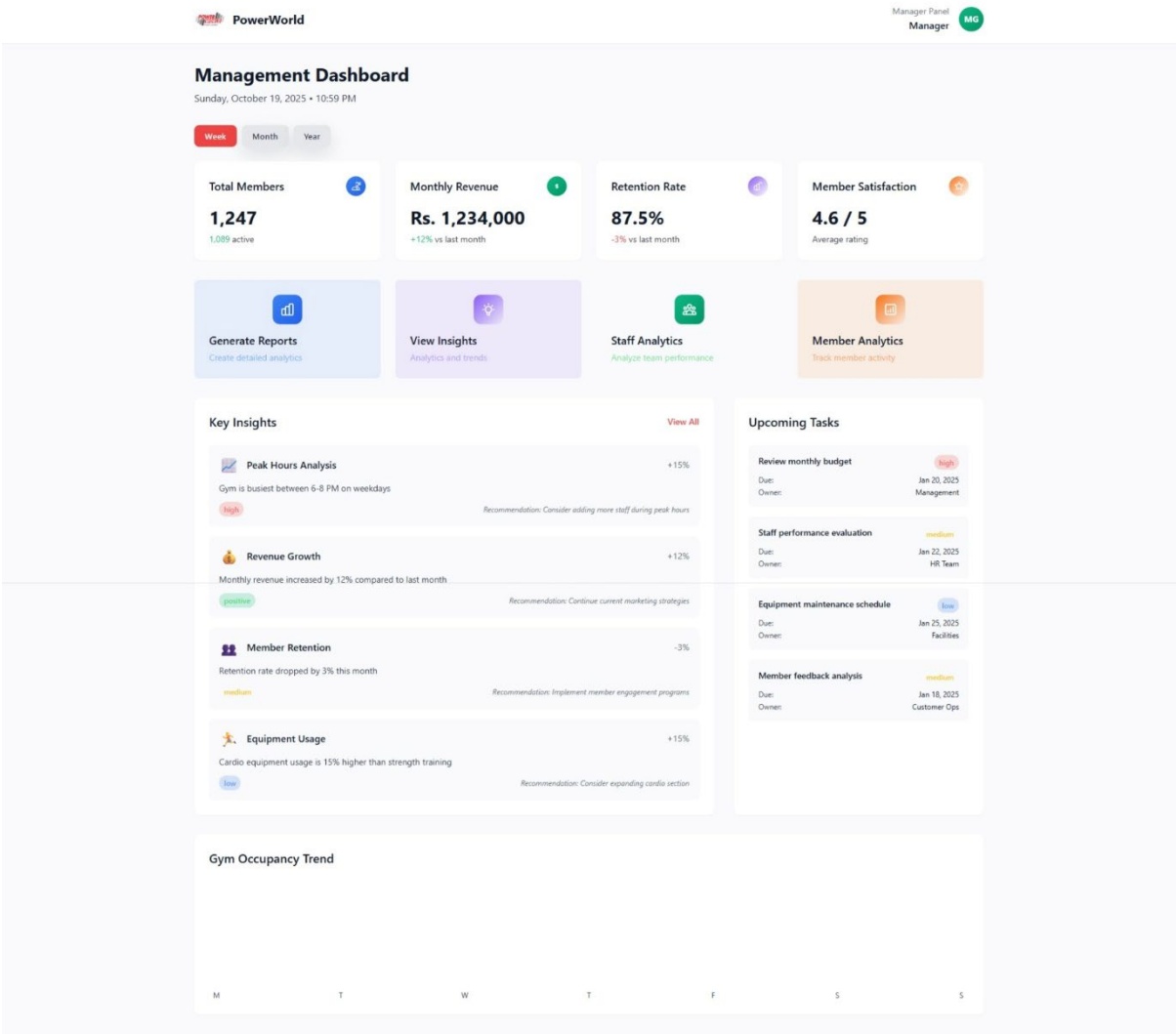


Figure 9: Mnager Dashboard

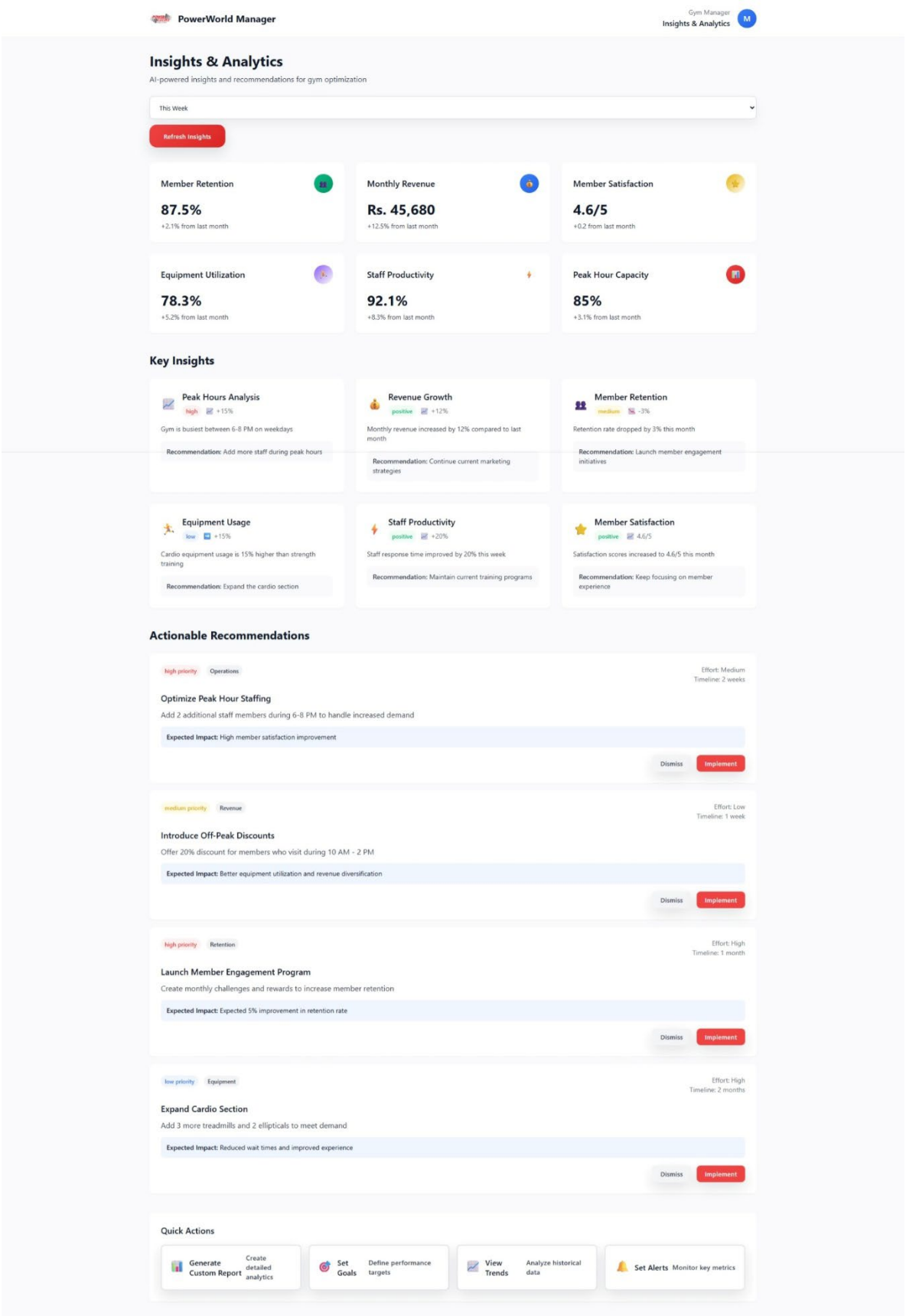


Figure 10: Mnager detailed insights

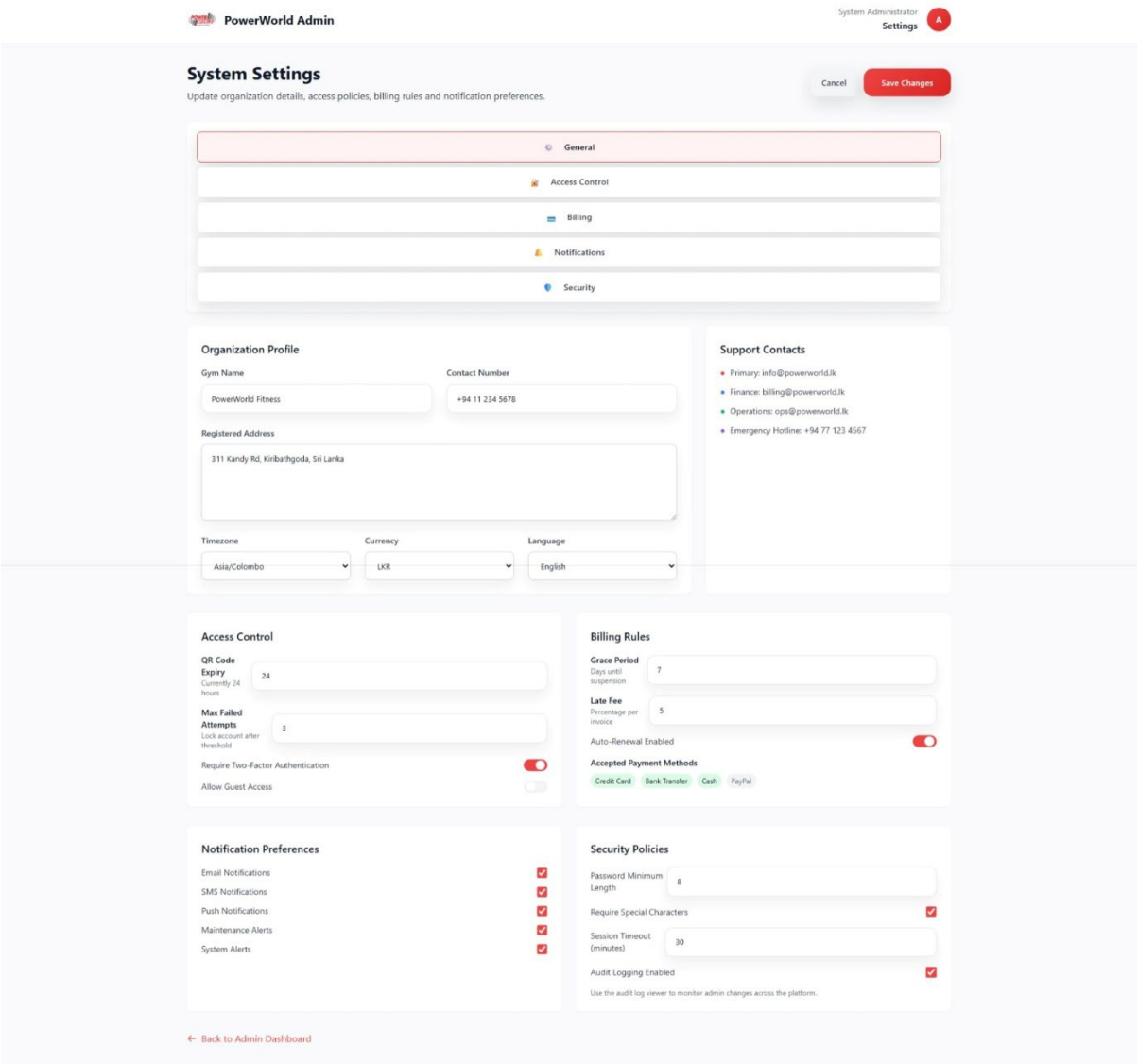


Figure 11: Configuring Admin settings

Conclusion

In conclusion, this documentation outlines the development of the **Gym Management Suite**, designed to overcome the challenges of the current manual system and enhance overall operational efficiency. By digitalizing key functions such as member registration, subscription validation, attendance tracking, and instructor management, the system ensures a smoother and more reliable workflow. The proposed solution focuses on improving accuracy, saving time, and providing a better experience for both members and staff. With well-defined requirements and a structured system design, the Gym Management Suite offers a modern, efficient, and user-friendly approach to managing gym operations.